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Review

Wire-Based Additive Manufacturing of Multi-Material Structures: A Review

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Abstract

Multi-material structures have great potential in high-end fields such as aerospace and energy. Which integrate the advantages of various metals and meet the demands of complex working conditions. Among additive manufacturing technologies for multi-material structures, wire-based processes have become a research hotspot due to their high material utilization, low cost, and high efficiency. This article systematically reviews the progress of research on this technology. The working principles and characteristics of common heat sources (WAAM, LWAM, EBAM) are introduced. Furthermore, the advantages and limitations of these heat sources for manufacturing multi-material structures are critically analyzed. Moreover, various metal wire combination systems (such as steel/Ni, Al/steel, Ti/Al, and Cu/Ti, etc.) were reviewed, and the differences and influences of different wire feeding methods and directions were discussed. The review highlights research findings on microstructure regulation, interfacial bonding mechanisms, and the mechanical property optimization of polymetallic structures. The influence laws of critical process parameters on structural properties are also elucidated. The existing problems in the current research were pointed out, and the future development trends were prospected. Unlike previous articles, this review establishes a more comprehensive process–structure–performance framework through the discussion of integrated heat source characteristics, wire feeding systems, and interface adjustment strategies. It aims to provide references for promoting the development and engineering application of additive manufacturing technology for wire-based multi-metal structures.

Keywords: additive manufacturing; multi-material; wire; interface regulation; process optimization; progress and outlook

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1. Introduction

Driven by the rapid development of contemporary industry, the service conditions of key components are increasingly showing an extreme and complex trend, generally requiring them to have highly integrated multi-functional coupling characteristics and

wide environmental adaptability. However, components composed of single homogeneous materials, produced via traditional manufacturing processes, are constrained by inherent limitations of isotropic material properties. This performance bottleneck hinders their ability to adequately meet the growing demand for diversified, high-performance integrated requirements [1–3]. Multi-Material Structures achieve the collaborative optimization and spatial gradient distribution of functions and performances at the material level by innovatively integrating the superior properties of heterogeneous materials, such as high strength, lightweight, high thermal conductivity, electrical conductivity, excellent corrosion resistance, and specific functional responses. Thus, it provides a breakthrough solution for meeting the requirements of complex and changeable service conditions [4,5]. Owing to its prominent advantages in performance tailorability, lightweighting capability, functional integration, and life-cycle cost-effectiveness, multi-material structure technology exhibits immense application potential and broad prospects in frontier industrial sectors such as aerospace, automotive manufacturing, advanced electronic packaging, nuclear energy equipment, and intelligent packaging. Especially in the aerospace field, lightweight aluminum alloys are combined with high-strength titanium-based or nickel-based superalloys for critical load-bearing and thermal management components. In the energy sector, applications include manufacturing corrosion-resistant coatings with functional gradients for nuclear reactors, as well as producing highly efficient heat exchangers that integrate corrosion resistance and thermal conductivity [6–8]. This technology has thus emerged as a pivotal research direction in advanced manufacturing and materials engineering [9–11].

Conventional multi-material fabrication technologies rooted in the fundamental thermo-mechanical coupling mechanism, such as diffusion welding, rolling composite, and casting, achieve macroscopic-scale material integration by precisely regulating thermodynamic parameters to drive interfacial metallurgical reactions or mechanical interlocking of heterogeneous materials [11,12]. Although such processes have high density and wide material adaptability, there are problems such as limited realization of functional gradients, insufficient forming ability of complex topological configurations, and low production efficiency caused by the series connection of multiple processes [13,14]. Over the past few years, the rapid advancement of AM technology has offered novel approaches to addressing the aforementioned issues. Through additive manufacturing technologies such as electric arcs, electron beams, and lasers, multiple different materials can be melted and deposited to fabricate multifunctional parts with multi-material structures. Additive manufacturing tends to form different types of materials at different positions, rather than homogeneous composite materials formed by mixing, melting, and depositing multiple materials. In comparison to conventional manufacturing processes, additive manufacturing exhibits distinct technical advantages in rapid near-net-shape fabrication, coupled with enhanced material utilization efficiency, reduced manufacturing costs, and shortened production lead times. Especially, it can achieve mold-free, a high degree of freedom, and customized forming of precision and complex parts, which is conducive to forming specific materials at the preset positions of the parts to obtain excellent local performance [15–17]. In addition, the thermal stress during the AM process is relatively low, allowing for a continuous transition of the composition gradient at the interface. The bimetallic parts formed in this way have good bonding strength [18].

Multi-material AM (MMAM) provides solutions for the manufacturing of cross-scale functionally integrated structures by realizing the precise regulation of heterogeneous material spatial distribution through digital material directional deposition. MMAM technology utilizes existing additive manufacturing processing methods to form components as a whole using a variety of materials with excellent performance, thereby improving the performance of components or realizing multiple special functions of parts [19]. The tech-

nology enables precise adjustment of material deposition, allowing different materials to be spatially allocated to any location within a component based on specific environmental conditions or structural requirements. This capability supports highly customized production and overcomes numerous limitations associated with conventional multi-material fabrication techniques. It is particularly suitable for manufacturing functionally graded and structurally complex parts, offering low production costs and excellent performance. Thus, it represents an ideal forming solution for high-performance components [20,21]. MMAM technology can form structures with complex material and geometric properties, as well as more functional solid structures, such as functional gradient material structures, composite structures, and bimetallic structure (BS). As the most promising technology, metal MMAM has made rapid development in recent years [22].

Metallic MMAM is usually a combination of properties, such as physical properties and biocompatibility of different materials, to achieve an extraordinary combination of properties, and the most commonly used metals include nickel-based alloys, copper alloys, titanium alloys, aluminum alloys, and stainless steel [23–27]. Metallic MMAM offers superior performance compared to single-material parts [28,29]. AM of metals can be categorized into powder and wire according to the feeding method. Furthermore, when powder is employed as the raw material, it tends to be more susceptible to the formation of defects such as porosity and cracking [30–32].

Furthermore, in comparison to powder-based AM processes, wire-based AM technology exhibits a significantly lower material cost. By reducing the complexity and surface quality of parts, the deposition rate is higher, and larger-sized parts can be obtained, as shown in Figure 1. The possibility of oxide contamination is also lower [33]. At present, the additive manufacturing technology of wire materials has been preliminarily studied and applied to the manufacturing of polymetallic parts [34].

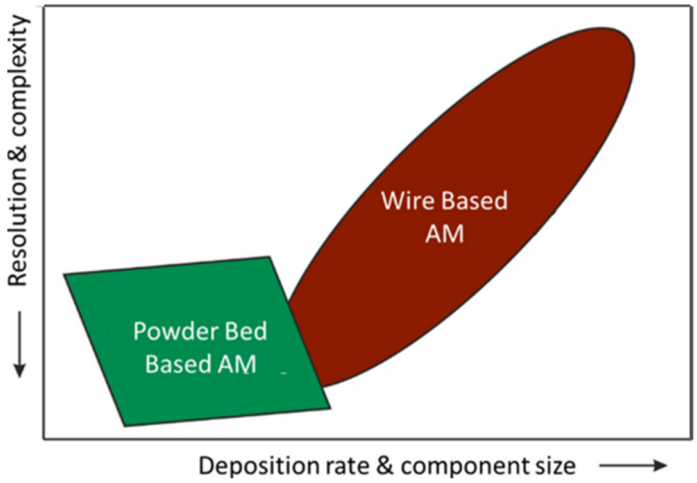


Figure 1. Comparison between powder bed-based and wire-based AM [33].

Drawing upon recent research advancements in metal MMAM, this paper presents a systematic review of the current state of the art regarding wire-based AM technology within the domain of metal heterogeneous material integration. The key problems faced by the MMAM technology of metals were summarized, and its development trend was predicted. To systematically sort out the technical system and research focus of multi-material wire additive manufacturing, this paper constructs the core framework diagram as shown in Figure 2. The technical background and key challenges of this field are comprehensively presented from the dimensions of material systems, process methods, and performance regulation.

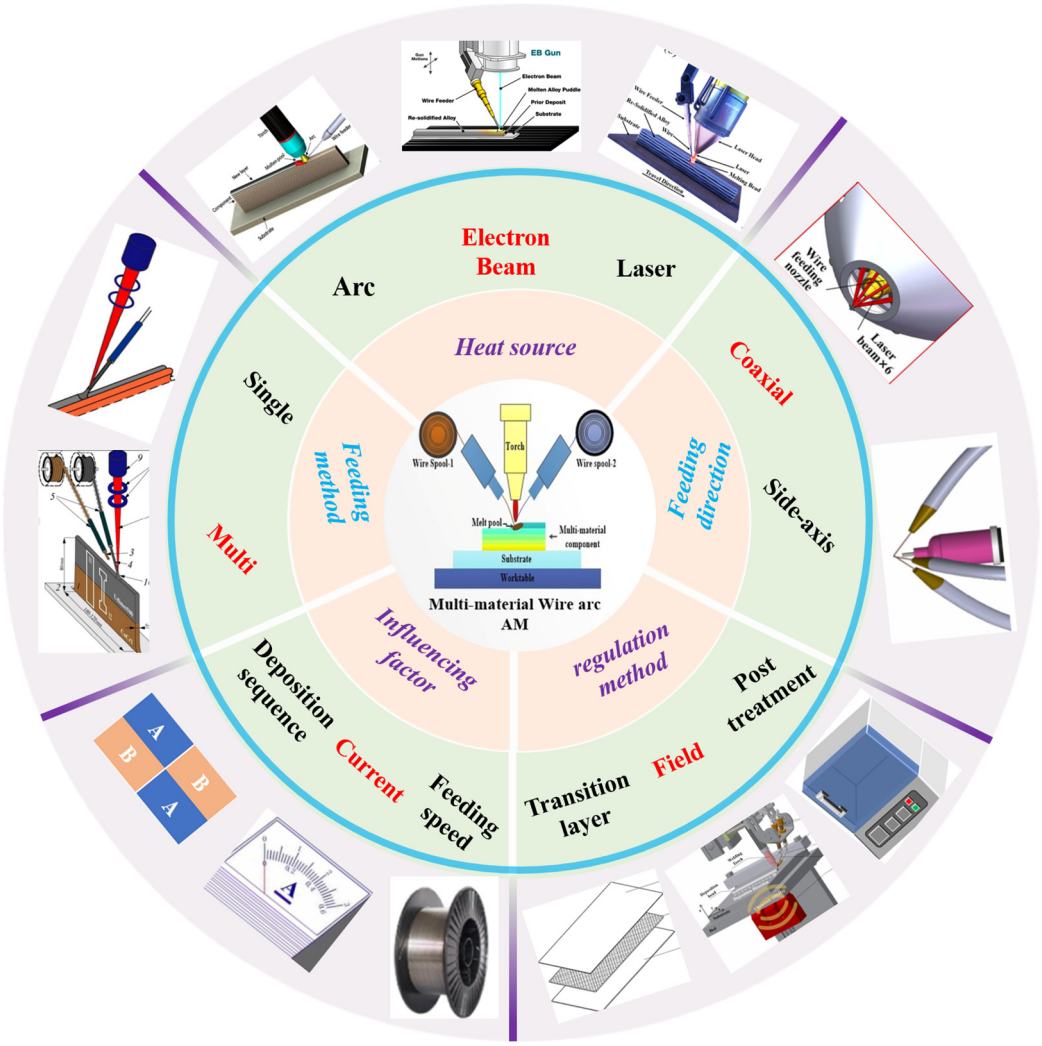


Figure 2. Overview of the core framework of wire-based MMAM.

2. Heat Source of Metal Wire Additive Manufacturing

Metal wire additive manufacturing technology, as an important branch of AM, is an advanced manufacturing technology that uses metal wire as the forming material. Through a specific energy source, the wire is melted and then stacked layer by layer along a preset path, ultimately achieving the formation of three-dimensional solid parts. Based on distinct energy source categories, metal wire-based additive manufacturing technology can be further categorized into three mainstream technological categories: arc-based wire-fed additive manufacturing, laser-based wire-fed additive manufacturing, and electron beam-based wire-fed additive manufacturing. According to the current research status, among the three technologies, wire arc additive manufacturing (WAAM) technology accounts for the largest proportion, approximately 70% to 80%, as shown in Figure 3.

Different technologies, with their unique energy transfer mechanisms, process characteristics, and material adaptability, have demonstrated significant advantages in their respective application scenarios, jointly promoting the development process of low-cost, high-efficiency, and multi-material integrated manufacturing of metal components. Table 1 presents a summary of various additive manufacturing methods applicable to wire materials.

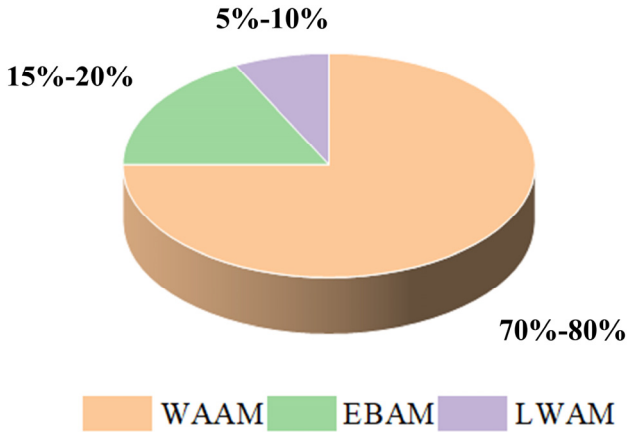


Figure 3. Proportion of additive manufacturing technology for metal wire materials.

Table 1. A summary of advantages, disadvantages, and applications of different heat sources.

Methods	Heat Sources	Advantages	Disadvantages	Applications	Ref.
WAAM	Arc	Low cost, high material utilization rate, fast deposition	Large thermal deformation, low precision, and numerous defects	Large and low-precision structural components	[35–37]
EBAM	Electron beam	High energy, vacuum anti-oxidation, small heat-affected zone	High cost, limited size, and high energy consumption	Precision components for aerospace and nuclear industries	[38,39]
LWAM	Laser	Low stress, high precision, stable quality	Slow deposition and poor compatibility with highly reflective materials	Complex components, medical implants, precision molds	[40–42]

2.1. Wire Arc Additive Manufacturing

WAAM is an advanced manufacturing technology rooted in the discrete-layer stacking manufacturing paradigm. Employing an electric arc as the heat source, it melts metal wires and deposits them layer by layer in accordance with a predefined path, thereby realizing the fabrication of three-dimensional solid components, as shown in Figure 4 [43]. WAAM technology has undergone a hundred years of development since the first patent was filed around 1925, and now it has become an important technological route for the fabrication of large metal components [44,45]. Boasting superior equipment and material cost-efficiency, material utilization rates approaching 100%, and excellent deposition efficiency, this technology enables the rapid realization of low-cost, high-efficiency fabrication of large, moderate-complexity aerospace structural components. However, its high heat input and deposition rate result in a large surface roughness of the molded component, which requires subsequent machining.

In recent years, significant progress has been made in process optimization such as multi-wire arc, composite additive manufacturing, pulse and vibration assistance, expansion of new alloy materials and metal matrix composites, intelligent process monitoring and feedback control, and application of digital twin technology, which have been widely applied in aerospace, energy, molds and other fields [46–49]. Eimer et al. [50] prepared Al/Zn bimetallic assemblies using the WAAM technique and investigated the effect of different process parameters and configurations on the process stability and microstructure of the deposited material, which resulted in high Zn concentrations without macroscopic segregation. Under good process conditions, bimetallic components can even achieve a strength higher than that of the base material. Wu et al. [51] fabricated steel/nickel structural components by a WAAM process and found that the tensile strength greatly exceeded that of nickel alloy or

stainless steel homogeneous material structures. Furthermore, the solid solution strengthening resulting from the interdiffusion of Fe and Ni enhanced the interfacial strength, exceeding the sum of the individual strengths of the constituent components.

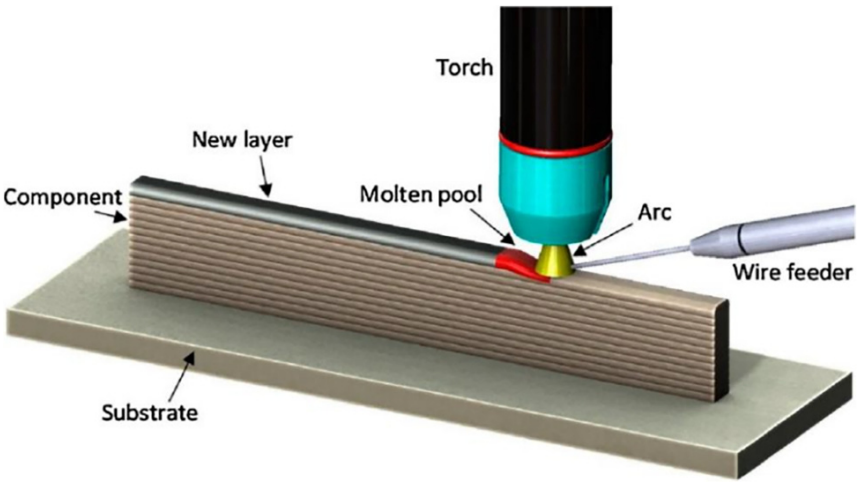


Figure 4. Schematic illustration of WAAM [43].

Similarly, Dharmendra et al. [52] fabricated mixed components of 316L steel and nickel-aluminum bronze using the WAAM process and found that the macroscopic interfaces of the samples were continuous and intact, without any fusion defects, pores, or cracks. Microstructure observations show that as copper penetrates along the cracks in the heat-affected zone, intermetallic compounds (IMCs) of Fe-Al are formed, further confirming that the interfacial bonding of the composite material is good and a good metallurgical bond is formed, which proves that WAAM has the ability to manufacture metal multi-material parts.

According to the different characteristics of the arc, WAAM can be further classified into three process types: plasma arc welding (PAW), gas tungsten arc welding (GTAW), and gas metal arc welding (GMAW) [53].

Among them, the GMAW technology uses wire as the melting electrode. In the GMAW system, the metal wire serves as the consumable electrode, and the welding torch is configured coaxially with the wire. The welding torch is manipulated by the welding robot system that executes the welding process, as shown in Figure 5. The welding wire operates as a self-consumable electrode. The arc generated between the welding wire and the workpiece melts the material of the welding wire and deposits the molten material on the surface of the substrate [54,55], as shown in Figure 5b.

Due to its high speed, there is no directionality issue in the forming process, and its efficiency is 2 to 3 times that of the GTAW and PAW methods. GMAW is the most widely used technology in WAAM and is particularly suitable for the manufacturing of large and complex components [37,56]. Reisgen et al. [57] adopted the dissimilar multi-wire WAAM method, by controlling the wire feeding speed (WFS) to adjust the composition ratio, gradient functional material components were prepared, and the relationship between the composition ratio and the WFS ratio was studied. For instance, Takeyuki et al. [58] studied the preparation of gradient materials by using double arcs and dissimilar double welding wires. Ni6082 and SS308L welding wires were, respectively, combined with two sets of GMAW equipment. The two arcs alternately ignited and deposited different welding wires, achieving alternating deposition of dissimilar welding wires and obtaining gradient functional material parts. In addition, Xiong et al. [59] employed the GMAW system and the method of auxiliary wire filling to regulate the composition of the deposited metal by adjusting the proportion of auxiliary wire filling, thereby preparing gradient materials.

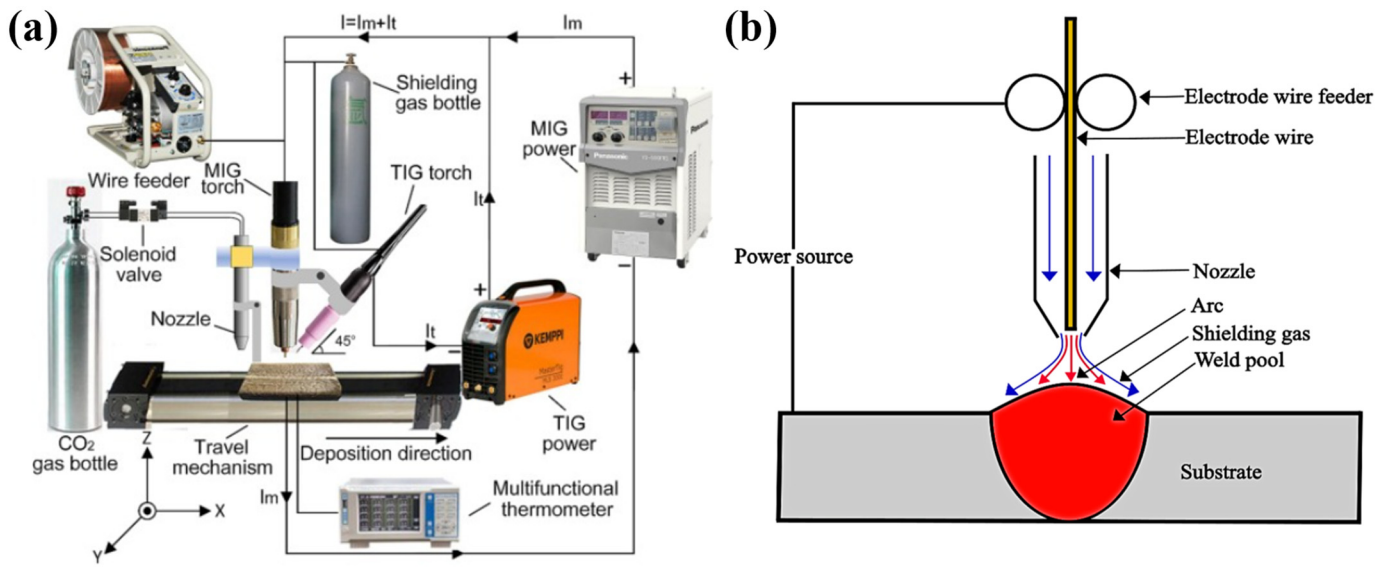


Figure 5. The system of GMAW. (**a**) Equipment diagram; (**b**) Schematic diagram [54,55].

It is worth mentioning cold metal transfer (CMT), which is characterized by dynamic wire feeding and is an advanced variant of traditional GMAW. It achieves droplet transition in a nearly zero current state through the combination of droplet short-circuit signal feedback and wire retraction mechanism, as shown in Figure 6 [60], which is conducive to precisely regulating the shape and microstructure characteristics of the deposited material [61,62]. This method enables the enhancement of dimensional accuracy and consistency of components, rendering it especially well-suited for high-precision-demanding applications. Motwani et al. [63] successfully prepared a thin-walled composite structure of nickel-based alloy and 316LSi stainless steel by the CMT process, and fabricated high-quality ultra-thin-walled deposits between austenitic stainless steel and nickel-based alloy. Tensile tests were conducted within the structure layer of 316LSi and alloy 625, and laterally at the interface of the titanium alloy. The average ultimate tensile strength (UTS) reached 660 MPa, and the elongation reached 49.3%. Karim et al. [64] fabricated 316L stainless steel/4043 aluminum heterogeneous composites via arc-directed energy deposition (DED) technology based on the CMT process, while investigating the influence of heat input on the morphology, porosity, and microstructure of the interfacial regions of the fabricated composites. Particular emphasis was given to the formation of IMCs. The results indicated that specimens fabricated under high heat input conditions exhibited a relatively thick IMC layer at the steel/aluminum interface, which led to increased interfacial brittleness and a deterioration in mechanical properties.

The difference between the GTAW and GMAW processes is the separation of the energy input from the filler material. As a result, GTAW allows for the simultaneous use of multiple wires. Compared to GMAW, the arc in GTAW is more stable, leading to uniform deposition and more accurate chemical composition [65], as shown in Figure 7a. In GTAW, an electric arc is generated between the non-consumable tungsten electrode and the substrate for heat generation, as shown in Figure 7b. The filler wire is side-fed and melted via arc heat, then deposited onto the substrate surface to attain the desired geometry and mechanical properties [54,66]. Baufeld et al. [67] deposited different metals such as 308 stainless steel, IN718 nickel-based alloy, and Ti6Al4V titanium alloy using GTAW, and analyzed the relationship between the mechanical properties and microstructure of different deposited parts. It was found that the UTS was between 929 MPa and 1014 MPa, depending on the direction and position of the tensile specimen. Tensile tests perpendicular

to the sedimentary layer showed a failure strain of 16 ± 3%, while tests parallel to the layer gave a lower value of approximately 9%.

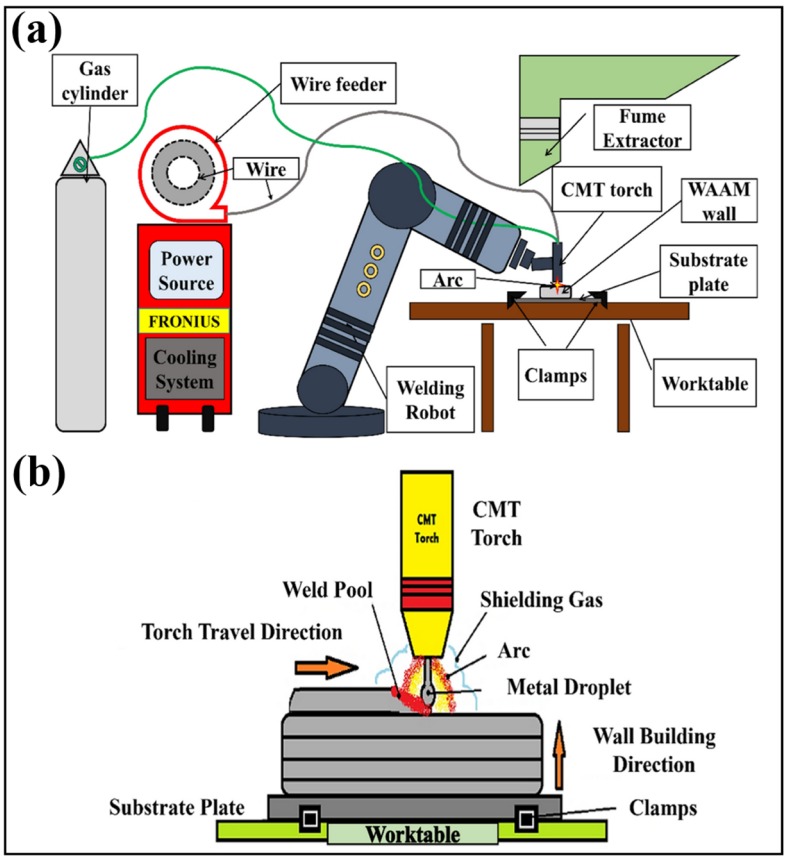


Figure 6. The system of CMT. (**a**) Equipment diagram; (**b**) Schematic diagram [60].

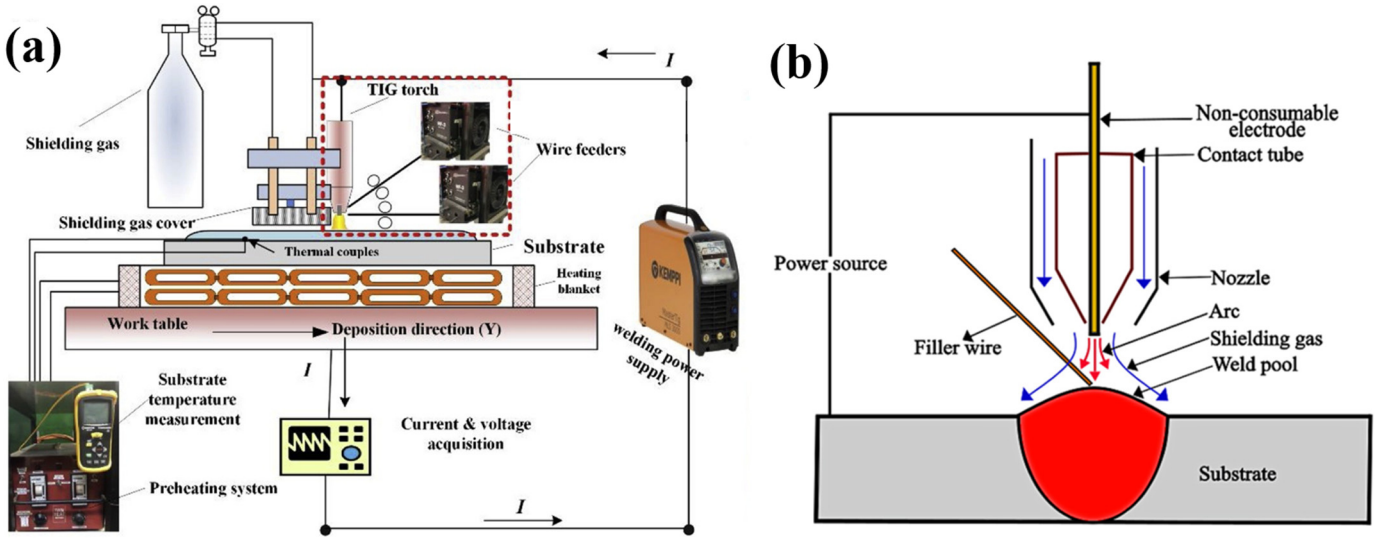


Figure 7. The system of GTAW. (**a**) Equipment diagram; (**b**) Schematic diagram [54,66].

Shen et al. [68] successfully obtained Al composite materials by feeding iron wire and aluminum wire, respectively, using GTAW arc and adjusting the wire feeding ratio. Along the bottom-to-top direction of the specimen, the hardness values increased from 150 HV to 650 HV, while the tensile strength increased from 39.5 MPa to 145 MPa. These results indicate that the composition gradient within the gradient region exerts a significant

influence on the microstructure and mechanical properties of additively manufactured components. Similarly, Wang et al. [66,69] achieved the gradient material of titanium-aluminum alloy by simultaneously feeding Ti6Al4V and Al wire through the method of arc additive manufacturing and introduced vanadium as a ternary alloying element. It was found that the introduction of vanadium, due to the general disappearance of the γ phase between dendrites and the toughening effect of V and significantly enhances the microhardness and tensile properties of the alloy.

PAW, on the other hand, boasts the highest energy density, enabling the high-speed forming of high-melting-point refractory metals and reducing deformation. The difference between PAW and GTAW lies in that the arc is restricted through a nozzle, generating a more concentrated and stable beam, as shown in Figure 8. This improves heat transfer efficiency and facilitates an increased welding speed [70]. Furthermore, PAW exhibits similarities to GTAW as both processes utilize non-consumable electrodes. However, PAW diverges from GTAW in terms of torch configuration, and owing to the confinement of the plasma arc within the nozzle, PAW demonstrates superior performance compared to GTAW. In PAW, an electric arc is generated between the tungsten electrode and the water-cooled nozzle, as shown in Figure 8b. The inert gas flowing through the arc area of the welding torch is ionized; that is, the gas becomes a plasma state. Then, the plasma jet is transmitted to the substrate through a small hole. The plasma jet transforms the filled welding wire into a molten form due to high heat. In addition, protective gas is used to protect the molten pool from contamination. However, compared with GMAW and GTAW, the initial cost of PAW is considerably higher [54,56].

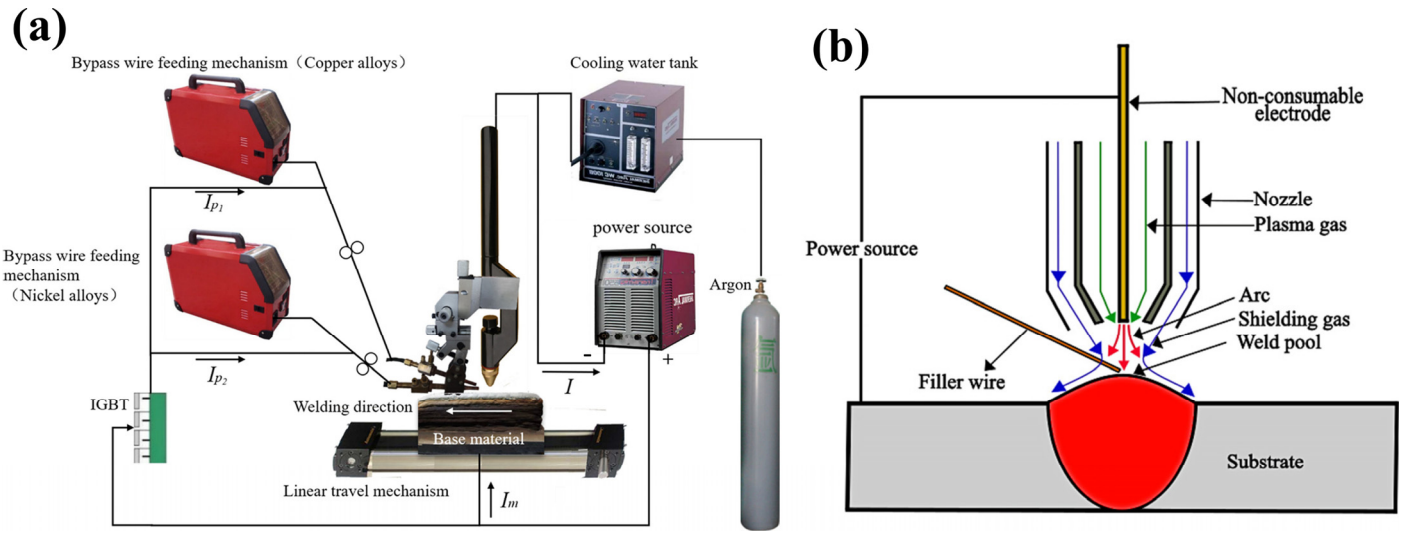


Figure 8. The system of PAW. (**a**) Equipment diagram; (**b**) Schematic diagram [54,56].

Miao et al. [56] fabricated Cu-Ni alloy gradient thin-walled structures via bypass-current PAW, and verified that the application of bypass current in the additive manufacturing of Cu-Ni gradient structures can enhance forming accuracy and effectively eliminate lateral compositional deviation within the gradient layers. Furthermore, within the Cu-Ni mixed gradient layers, the average grain size of the gradient layers decreases as nickel content increases, while abnormally large secondary grains are observed across different gradient layers. Owing to variations in Ni content, varying degrees of amplitude modulation decomposition occur in different gradient layers, leading to alterations to grain orientation and texture composition. Under the combined regulation of post-treatment, results indicate that the microhardness and tensile strength of the alloy exhibit a trend of first increasing and then decreasing as nickel content increases, peaking on the side

where the composition gradient is biased toward nickel. Ayan et al. [71] fabricated low-alloy steel/stainless steel gradient metallic materials via the utilization of low-alloy steel (ER70S-6) and austenitic stainless steel (308LSi) metallic wires. No defects were detected at the functionally graded material (FGM) interface, and the tensile strength exhibited an increase of up to 46% compared to single-material fabrication. In hardness measurements and microstructural investigations, the FGM structure was observed to exhibit varying properties across the gradient layers. The findings indicate that the fatigue limit of the FGM component in the horizontal direction is 25% higher than that in the vertical direction.

Through the above research, the capabilities of WAAM technology in the manufacturing of metal multi-material parts have been fully verified, ranging from the detailed applications of different process types, such as GMAW, GTAW, to PAW, to the successful preparation of gradient functional materials. In the future, with the continuous innovation and improvement of technology, WAAM technology is bound to play a greater role in more high-end manufacturing fields and inject strong impetus into the development of advanced manufacturing industries.

2.2. Laser Wire Additive Manufacturing

Laser wire additive manufacturing (LWAM) is an advanced additive manufacturing technology developed based on the principle of laser cladding. The LWAM systems are typically composed of an axis robot, a laser head, an automatic wire feeding system, and occasionally a positioning stage, as shown in Figure 9a. To mitigate contamination during the deposition process, LWAM systems typically employ protective gas shielding. When processing active metals, it is imperative to utilize a fully inert chamber equipped with adequate gas supply and sufficient purge time to attain the required oxygen concentration [38]. The core principle is to use a high-energy-density laser beam to melt the synchronously conveyed metal wire material, and to create a molten pool by using the metal wire raw material and the laser. The gas shield is used to keep the deposited material from being contaminated. This process involves melting metal wires with an energy source to form liquid droplets, and then adding the liquid droplets layer by layer to produce the final product, as shown in Figure 9b. This technology combines the advantages of the high precision of laser processing and the high efficiency of wire feeding, and shows unique value in aerospace, high-end molds, medical implants, and other fields [38].

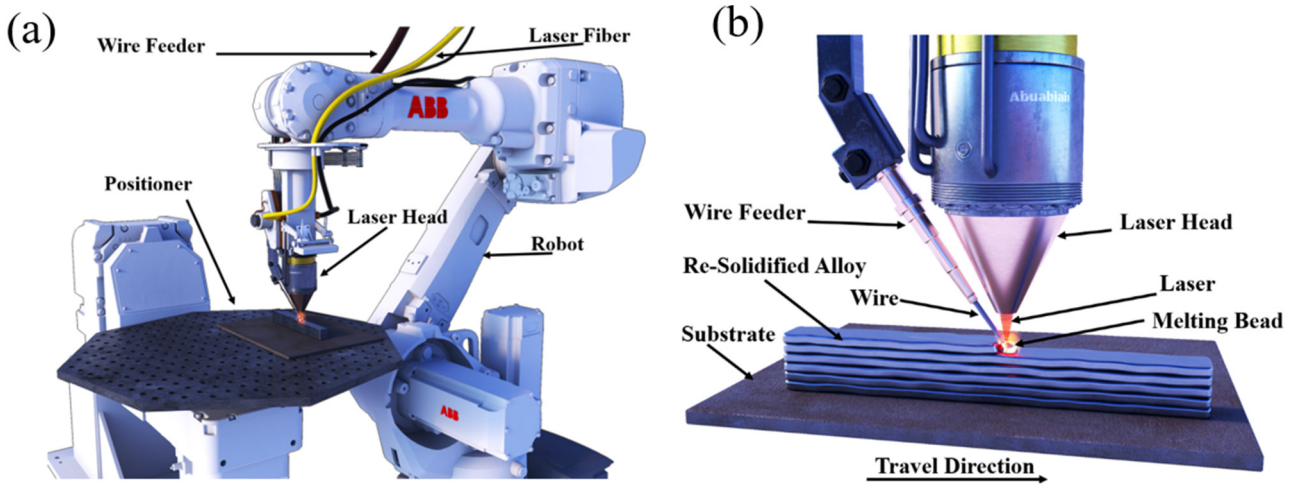


Figure 9. The system of LWAW. (**a**) Equipment schematic diagram; (**b**) Schematic illustration of the LWAM bead deposition mechanism [38].

The key to the process control of LWAM lies in the precise matching of parameters such as laser power, WFS, and height increment. Research shows that in laser wire feeding

additive manufacturing, the width of the surfacing layer mainly depends on the laser power, while the height of the surface layer is closely related to the welding speed, and the cross-sectional area of the weld seam is mainly controlled by the wire feeding rate.

In addition, the direction of wire feeding has a significant impact on the forming quality [72]. Harwell et al. [73] developed a laser metal deposition system incorporating coaxial wire feeding. This configuration ensures that the shielding gas, filler wire, and laser beam converge at a single focal point, thereby optimizing energy coupling and enhancing process stability during interaction between the laser head and the workpiece. Zapata et al. [74] investigated the influence of coaxial laser wire deposition process parameters using aluminum and stainless steel wires on the wire head and layer geometry. Insufficient energy input may lead to discontinuous deposition or unmelted wire beads, whereas excessive energy input, insufficient WFS, or inadequate focal offset may result in the premature melting of the wire prior to its arrival at the substrate surface. They found that the relationship between process parameters and the height and width of the weld head can be approximated by a linear model. In addition, Huang et al. [75] investigated the correlation between process parameters and bead geometry for aluminum and its alloys fabricated via LWAM. They developed a second-order polynomial model to correlate LWAM process parameters with layer geometric characteristics and identified that the maximum deposition weight was attained at a wire feeding angle of 45°. Additionally, they examined the influence of travel speed and laser power on deposition width, revealing that laser power exerted a more significant effect than travel speed.

Syed et al. [76] developed a hybrid process integrating coaxial powder feeding and lateral wire feeding, under which the chemical composition of the component structure remains consistent. This hybrid process was found to enhance deposition efficiency and reduce surface roughness. While the specimens fabricated via this method exhibit some porosity, the porosity is 20–30% lower than that of powder-only processes. Additionally, the spinneret angles in both horizontal and vertical directions play a critical role in achieving high deposition efficiency and superior surface quality. In addition, the position of the laser focus is equally important. Motta et al. [77] conducted coaxial LWAM using a multi-module fiber laser and analyzed the influence of the focus position on part forming during the additive manufacturing process. When the focal length in Figure 10 is too large, the melting is insufficient. When the focal length is too small, excessive melting occurs, resulting in dripping. Both can easily cause melting defects in the welding wire, indicating that an appropriate focal position is conducive to improving process stability.

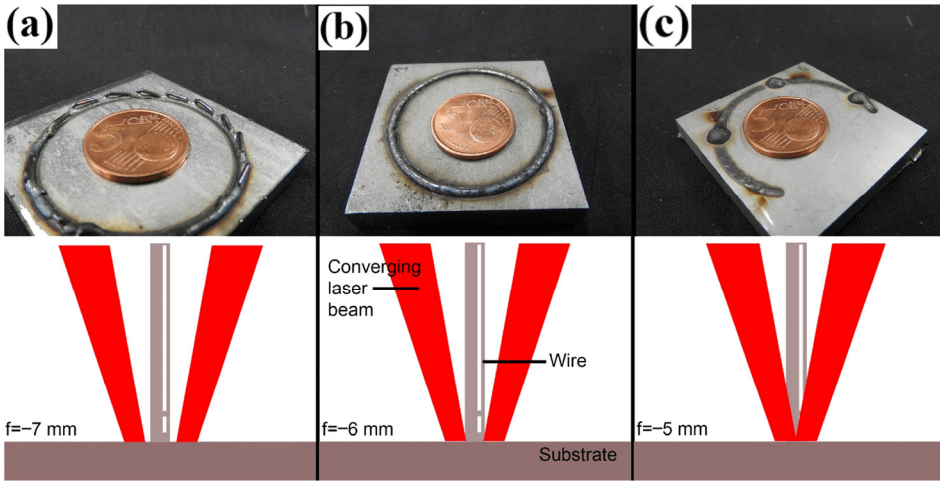


Figure 10. The effect of focal position on the layer adherence stability (**a**) −7 mm; (**b**) −6 mm; (**c**) −5 mm [77].

In addition, process compounding is also a current research hotspot. Laser-arc hybrid manufacturing is a novel manufacturing technology that integrates laser and arc heat sources, whose manufacturing principle is illustrated in Figure 11a. Specifically, the interaction mechanism between laser and arc is as follows: Arc heating improves laser energy utilization efficiency, while the introduction of laser exerts stabilizing and arc-inducing effects [78]. This hybrid manufacturing technology typically finds applications in shipbuilding, transportation equipment manufacturing, and pipeline fabrication [79]. Gong et al. [80] proposed a laser-oscillating arc AM approach. The surface roughness of the fabricated specimens was reduced to 20% of that achieved via arc additive manufacturing, while porosity was effectively mitigated. Furthermore, uniform tensile properties and minimal tensile anisotropy were realized. The morphology of the deposited surface and the sidewall accuracy are illustrated in Figure 11b.

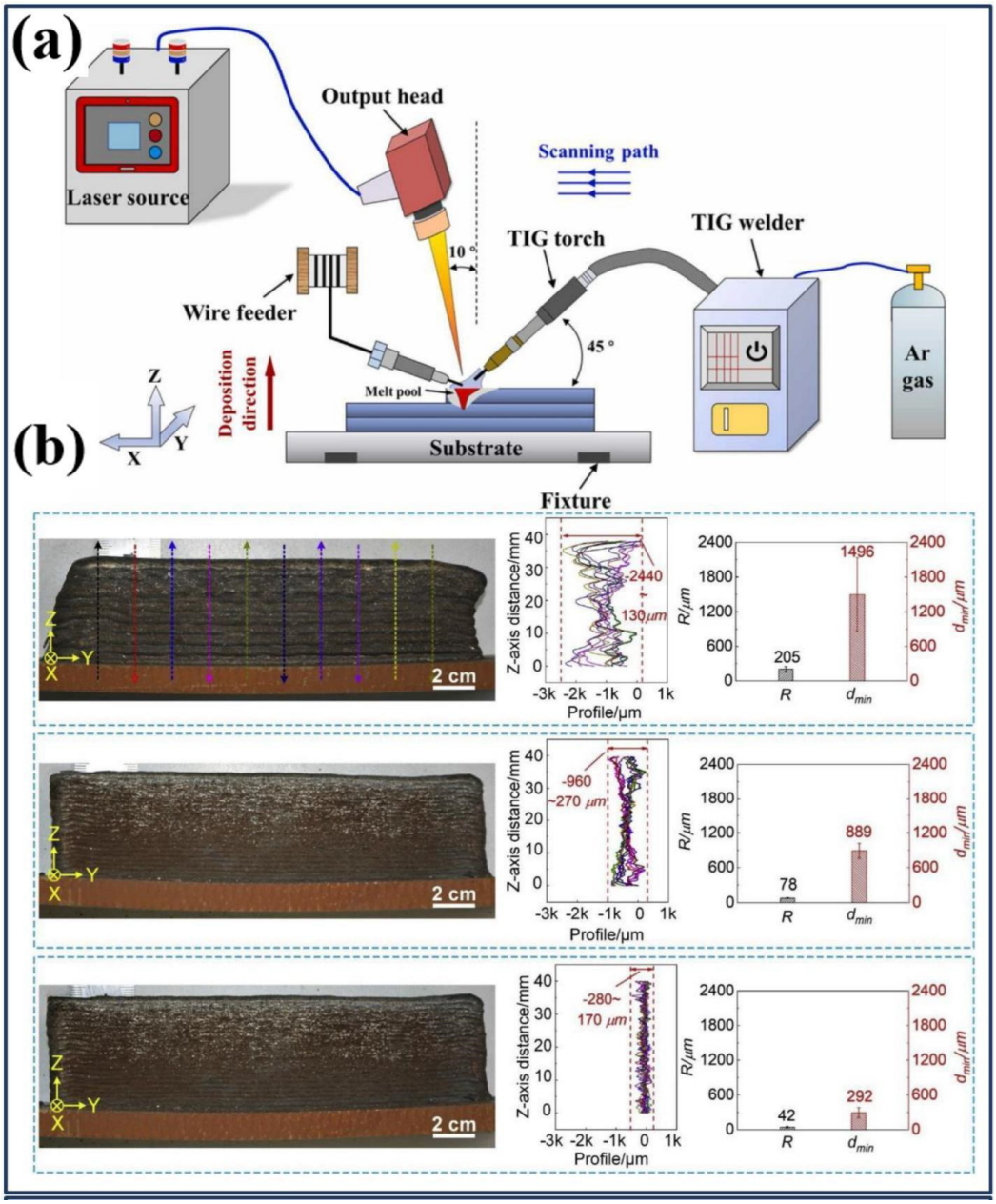


Figure 11. (**a**) Schematic illustration of laser–arc hybrid additive manufacturing [78]; (**b**) Deposited surface profile and sidewall surface accuracy [80].

Laser wire feeding AM technology, with its high precision, high efficiency, and multi-material adaptability, is accelerating its transformation from laboratory research to industrial mass production. Compared with WAAM, LWAM has higher energy focusing and a smaller heat-affected zone, enabling more precise geometric feature control. Especially in the application of multi-material fields, it has broken through the limitations of traditional manufacturing technologies. By precisely controlling the laser energy and the coordinated feeding of multiple wires, the efficient integration of FGM, dissimilar metal connections, and composite material preparation has been achieved [81].

2.3. Electron Beam Additive Manufacturing

Electron beam additive manufacturing (EBAM) achieves deposition and shaping by melting metal wires with high-energy electron beams in a vacuum environment. The core advantage of this technology lies in its immunity to electromagnetic interference, extremely high energy utilization rate, and its suitability for the manufacturing of active metals, as shown in Figure 12 [82]. The EBAM incorporates a vacuum environment to mitigate electron scattering, atmospheric contamination, and oxide formation during the 3D printing process [83,84]. Electron beam wire-feed additive manufacturing enables the fabrication of components composed of various metals and alloys, exhibiting diverse overall dimensions, mechanical properties, and performance characteristics [85].

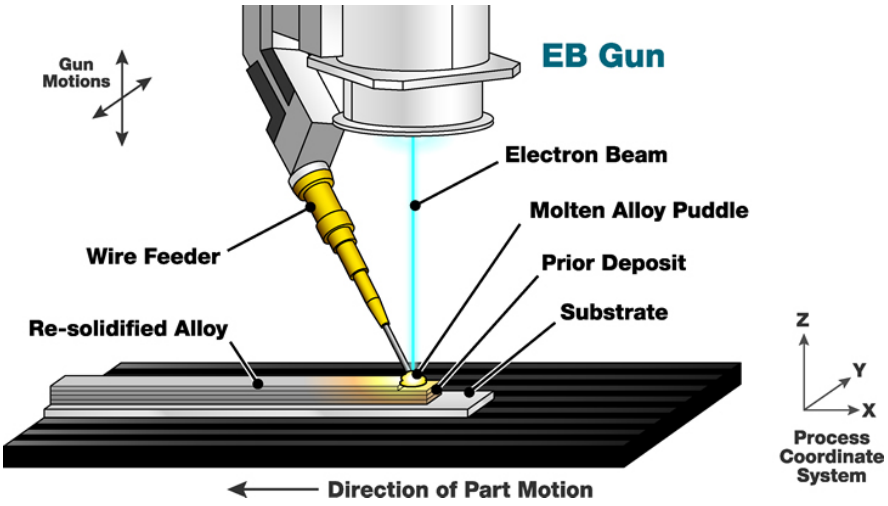


Figure 12. Schematic illustration of wire-based EBAM process [82].

In the application of multi-material fields, through the coordinated control of high-energy electron beams and multi-filament materials, the material limitations of traditional welding and additive manufacturing have been broken through. The electron beam 3D printing technology possesses at least two technical characteristics that ensure the microstructure uniformity of the fabricated specimens in the bonding zone. Specifically, the first is that the deposition process occurs in a vacuum environment, and the second is the utilization of a high-power electron beam. Under such conditions, the formation of oxides and other impurities is effectively these impurities would otherwise detrimentally affect surface quality and, consequently, the mechanical properties of the resultant polymetallic components [86]. Osipovich et al. [86] used double-wire EBAM to fabricate steel-copper bimetallic samples, as shown in Figure 13 [42]. The steel wire was fed into the molten pool, facilitating layer-by-layer metal deposition and enabling the fabrication of a complete 10-layer steel wall, as shown in Figure 13a. Upon reaching this number of steel layers, the copper wire was introduced into the molten pool via a secondary wire feeder. The gradient structure was formed by progressively increasing the copper wire feed rate (Vc) while

concurrently reducing the steel wire feed rate (Vs) until the steel wire feed was completely halted, as illustrated in Figure 13b. Through the analysis of the low-magnification structure of the cross-section of the steel-copper composite wall, it was found that the interlayer deposition of the composite wall was complete, without defects such as cracks and fissures. Moreover, the characteristics of the gradient zone structure with smooth changes in component concentration were revealed through microstructure analysis.

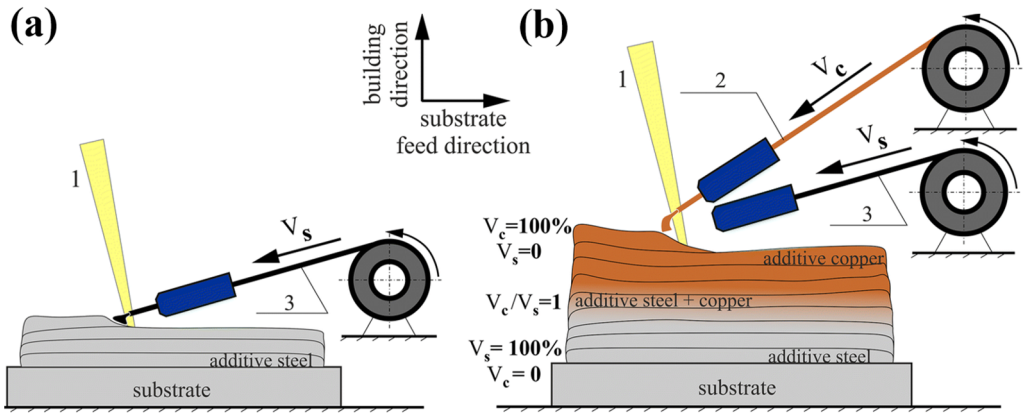


Figure 13. Gradient transition zone structure in steel-copper specimens fabricated via dual-wire-feed EBAM (**a**) Steel wall; (**b**) Copper-steel wall [42].

In addition, Utyaganova et al. [87] employed EBAM to fabricate AA7075/AA5356 gradient specimens, and observed that the chemical composition of the transition zone was characterized by a gradual increase in Mg content and a corresponding decrease in Cu and Zn contents. In terms of microstructure, it transitions from a coarse network of IMCs at grain boundaries to uniformly distributed IMCs particles. The phase constituent of this region comprises coarse and fine Al-Mg-Cu-Zn particles, along with Fe-Al particles and Mg_2Si intermetallic particles distributed at the columnar grain boundaries. The microhardness and tensile strength of this region exhibit higher values than those of the pure AA5356 deposited material. The average tensile strength attains 289 MPa, with an elongation at break of 21%.

Although the vacuum environment requirements of the EBAM system increase the complexity and cost of the equipment, they also bring unique advantages: on the one hand, they effectively prevent the oxidation of active metals; On the other hand, the dynamic behavior of the molten pool under vacuum conditions is significantly different from that under normal pressure, which is conducive to achieving deeper penetration and more uniform elemental distribution. This characteristic makes EBAM of special value in multi-material manufacturing, especially for the combination of easily oxidized metals.

As shown in Table 2, the three technologies of WAAM, LWAM, and EBAM exhibit significant performance differences in multi-material manufacturing. This comparison not only clarifies the applicable scenarios of each technology (for example, WAAM is suitable for large-sized and low-cost components, and EBAM is suitable for active metals) but also provides a quantitative basis for subsequent process selection and optimization.

Table 2. Comparison of key performance indicators of WAAM, LWAM, and EBAM technologies.

Technology/Indicators	WAAM	LWAM	EBAM	Ref.
Deposition rate (kg/h)	2–10	0.5–2	1–5	[33,38,48,65]
Deposition size	Very large	Medium to large	Large (limited by the vacuum chamber)	[33,46]
Forming accuracy	Lower (several hundred micrometers)	Higher (tens of micrometers)	Medium to high	[48,65,76]
Heat input	High	Low-medium	High	[46,88]
Cost	Low (based on mature welding equipment)	High (laser, robot system)	Very high (vacuum system, electron gun, maintenance)	[33,46]
Vacuum requirement	No (protective gas is usually required)	No (protective gas is usually required)	Yes	[76,78]
Scope of applicable materials	Extensive (steel, nickel-based, aluminum, copper alloys, etc.)	Extensive	Very extensive, especially proficient in active metals (Ti, Ta, Zr, etc.) and high-melting-point materials	[76,80]
Multi-material forming capability	Excellent (the multi-wire synchronous feeding technology is mature	Good (coaxial high wire feeding accuracy)	Good (double wire feeding can be achieved)	[86,89]

3. Wire Feeding System of Metal Wire Additive Manufacturing

3.1. Wire Feeding Method

The wire feeding method, as a core process link in the additive manufacturing of metal wires, directly determines the stability of material transportation, the control ability of component composition, and the forming efficiency. Through the design of the driving mechanism and the conveying path, it precisely feeds the metal wire into the molten pool, which is the key guarantee for achieving the layer-by-layer accumulation and forming of materials. Different wire feeding methods have significant differences in structural complexity, flexibility of performance control, and application scenarios.

3.1.1. Single-Wire Feeding

The metal wire is driven by a single-wire feeding wheel or two driving wheels and conveyed to the welding nozzle through the wire guide pipe, where it interacts with the arc heat source to form a molten pool. Compared with the multi-wire feeding method, the single-wire feeding system has a relatively simple structure, lower equipment cost, and the stability of wire feeding is easier to ensure, which is conducive to improving the dimensional accuracy of the formed parts. It can be used to prepare FGM components and has important applications in aerospace, energy, and other fields. Gürol et al. [90] adopted WAAM and directly deposited ferritic-austenitic BS on the substrate through single wire feeding. The experimental setup is shown in Figure 14a. Figure 14b shows the deposition strategy. An austenitic steel wire with approximately 17 wt.% Cr and 11 wt.% Ni was utilized for the deposition of the initial 30 layers. Subsequently, a metal-cored wire was adopted to deposit the final 30 layers on the steel deposited layer. Eventually, a total of 60 layers were deposited, yielding a bimetallic thick-walled structure. Meanwhile, according to the report by Turgut et al. [91], the interlayer residence time can significantly control the interlayer temperature, thereby modifying the mechanical and microstructural properties of WAAM components fabricated from low-carbon and low-alloy welding wires. Therefore, the author used an interlayer residence time of 180 s between consecutive layers to construct thermoplastic components, which helps to avoid heat accumulation during the subsequent deposition of molten metal.

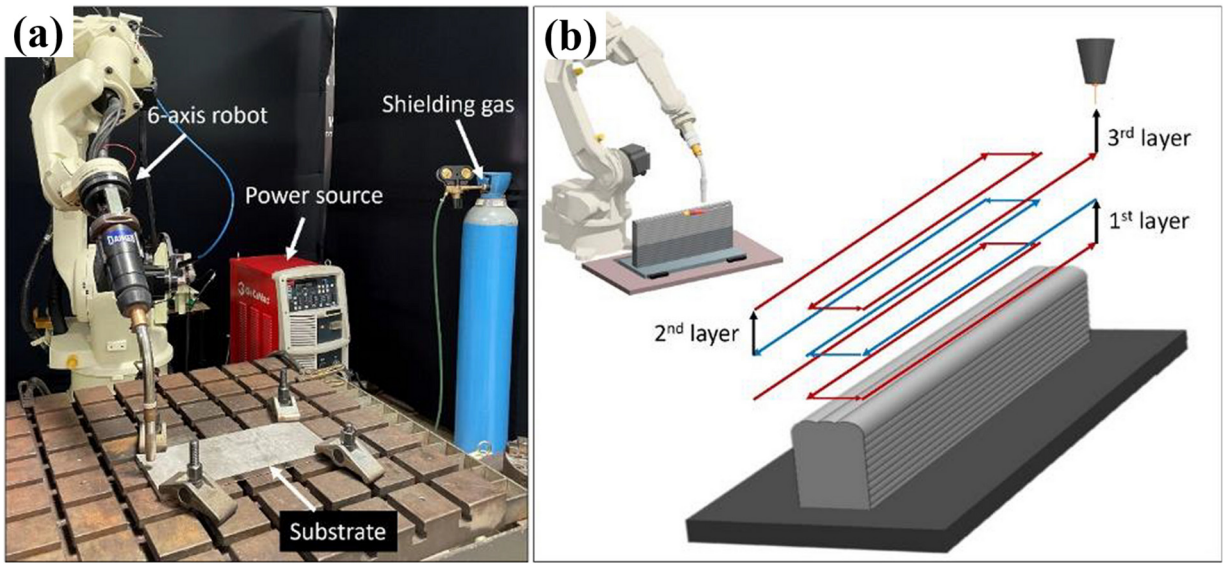


Figure 14. The system of (**a**) WAAM setup and (**b**) Deposition strategy [90].

In addition, Zhai et al. [92] directly deposited ER70S-6/S355 alloy components on the substrate by single-wire feeding arc additive manufacturing. There was a good metallurgical bond at the interface, and no defects were formed. After quenching heat treatment, the yield strength at the joint of ER70S-6/S355 manufactured by WAAM was 511.6 MPa, that of UTS was 740.7 MPa, and the elongation was 20.1%.

Monofilament feeding additive manufacturing has some limitations in application. In terms of materials, it can only use single-component filaments, and it is difficult to flexibly optimize local performance. In terms of process, the range of heat input control is limited, and it is not as flexible as the multi-wire process in controlling the bimetallic interface. In addition, the subsequent machining costs are relatively high. These limitations restrict its application in high-end manufacturing scenarios, and it is necessary to upgrade and break through by combining technologies such as multi-filament collaboration.

3.1.2. Multi-Wire Feeding

The multi-wire feeding technology in WAAM refers to the process of integrating multiple independent and controllable wire feeding mechanisms in a single additive manufacturing process to synchronously or alternately feed two or more metal wires of different compositions into the same molten pool along preset paths and proportions, thereby breaking through the limitations of a single material. Realize the precise customization or continuous gradient variation in material composition, microstructure, and properties at any position within the component [89]. Pattanayak et al. [93] fabricated ER70S-6 and SS316L alloy materials by multi-wire arc additive manufacturing (MWAAM). During the deposition process, the WFS of SS316L was adjusted between 0 and 1.5 m/min, while the WFS of ER70S-6 was fixed at 2 m/min. The feed rates of the two filler wires were monitored and controlled by the autonomous wire feeding system. There were no obvious defects in their macroscopic morphology. Microscopic elemental analysis was conducted on them, and it was found that the contents of Cr and Ni gradually increased with the increase in WFS-2. The anisotropy of the mechanical property test was <5%, indicating that an isotropic iron-based alloy was prepared. Zhou et al. [94] prepared the Al-Cu/Al-Mg-Sc bimetallic alloy by direct energy deposition with double-wire arc, as shown in Figure 15, and constructed a fine/coarse equiaxed crystal/columnar crystal triple heterostructure. This research guides the preparation and engineering application of heterogeneous microstructure alloys with different grain morphologies and various length scales.

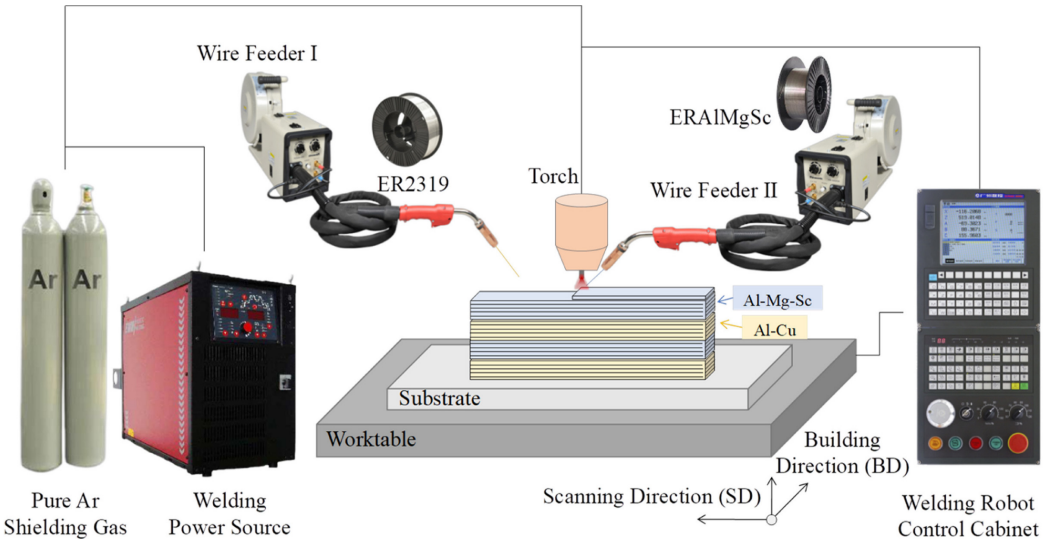


Figure 15. Schematic illustration of the dual-wire Arc-DED system [94].

Inspired by the biological gradient structure, Jiang et al. [95] used MWAAM technology. They observed that TC11 and TC4 titanium alloys underwent intermixing as a result of dilution, remelting, and convective mixing within the molten pool. During the transition between the two distinct titanium alloys, long-range gradient concentration distributions of alloying elements were formed. The surface tension gradient of liquid metal facilitates the dilution of alloying elements during the MWAAM process. During the deposition of TC4 alloy onto TC11 alloy, the interlaminar microhardness exhibits a gradual decrease. The average tensile strength and yield strength of the MWAAM titanium-based gradient heterogeneous alloy are 793.14 MPa and 770.89 MPa, respectively. Additionally, the average tensile strength of this titanium-based gradient heterogeneous alloy is close to that of the MWAAM TC4 alloy and attains approximately 85% of that of the MWAAM TC11 alloy.

To more clearly demonstrate the advantages of multi-wire feeding, Feng et al. [96] incorporated a dual-wire feeding mechanism into the plasma arc additive manufacturing process and conducted a comparative study with a single-wire feeding mechanism. The results showed that the dual-wire feeding mechanism exhibited a 0.06-fold higher deposition rate than the latter. Furthermore, the tensile strength and elongation were also enhanced by 10.2% and 176%, respectively.

Multi-wire feeding has significant advantages over single-wire feeding: by adjusting the feeding speed ratio of different wire materials, continuous gradient changes in composition can be achieved, allowing for the direct preparation of multi-metal composite components and gradient functional materials, and flexibly optimizing local performance. The heat input control is more flexible, which can reduce heat accumulation and deformation, enhance the stability of droplet transition and form accuracy, and improve the bonding quality of the bimetallic interface. The deposition efficiency can also be enhanced through multi-arc coordination, reducing the subsequent processing costs of complex components and better meeting the performance and efficiency requirements of high-end manufacturing.

3.2. Wire-Feeding Direction

The direction of wire feeding focuses on the spatial path design of the metal wire entering the molten pool, which directly affects the uniformity of wire melting, the distribution of heat input, and the interface bonding quality of dissimilar materials. By optimizing the relative position between the wire feeding channel and the molten pool, precise control of material transportation can be achieved, which is an important technical dimension

for improving the forming accuracy of components and expanding the multi-material composite capability.

3.2.1. Coaxial Wire Feeding

Multi-material coaxial wire feeding technology is an advanced additive manufacturing method. It uses coaxial nozzles that integrate multiple wire feeding channels to simultaneously or as needed feed two or more different types of metal wires during the manufacturing process. It uses electric arcs, lasers, and electron beams as heat sources to melt and layer by layer deposit them into shape. The core advantage of this technology lies in its ability to achieve seamless transition and composite of dissimilar metal materials (such as steel-aluminum, copper-steel, titanium alloy-nickel-based alloy) or FGM within a single component, breaking through the limitations of traditional single-material manufacturing and thus integrating specific properties of different regions on a single part.

Meeting the customized demands for complex multi-functional components in fields such as aerospace and energy equipment is one of the key technical directions for enhancing the design freedom and performance potential of additive manufacturing [97]. Yao et al. [98] deposited high-toughness and high-strength TC11 titanium alloy by coaxial wire feeding and multi-beam laser directional energy deposition. By optimizing parameters such as WFS, the formability was improved, as shown in Figure 16. This laser additive manufacturing system is equipped with a 1200 W fiber laser, which is divided into six laser beams, a coaxial wire feeding nozzle, a wire straightening machine, a high-speed camera, argon gas blowing, protective gas, and a six-axis linkage control system. The coaxial wire-feed laser directed energy deposition (WLDED) head in this operation can provide a large amount of uniform argon blowing protective gas to significantly reduce oxidation.

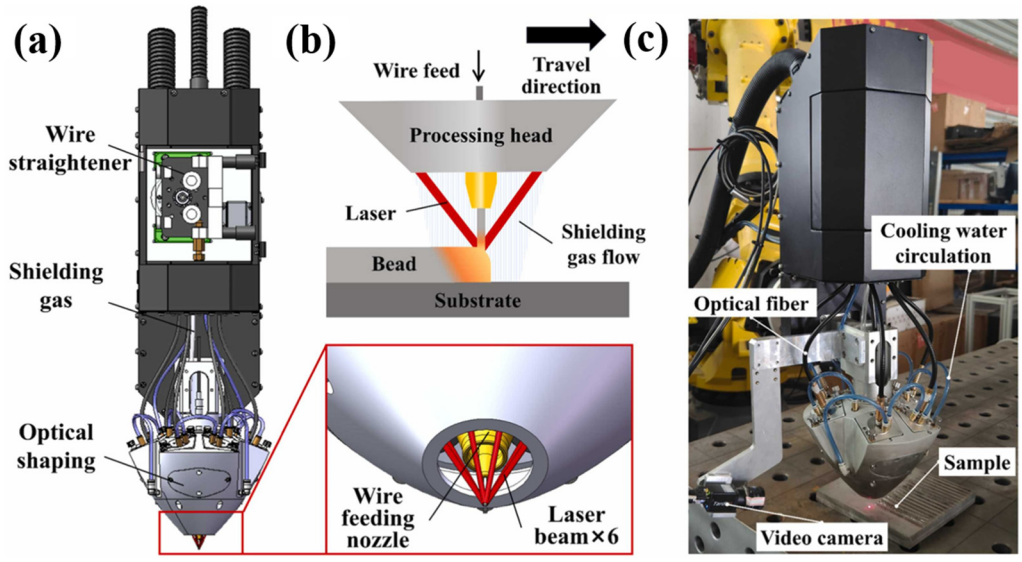


Figure 16. Proposed coaxial WLDED method: (**a**) Custom-designed coaxial processing head integrated with six laser beams; (**b**) Schematic diagram of the deposition process; (**c**) Fabricated processing head [98].

It was found that utilizing coaxial WLDED, which exhibits specific thermal gradient and solidification rate characteristics at a specific laser-wire coupling position, could significantly influence microstructural evolution in different regions and mechanical properties along typical orientations. Additionally, the synergistic effect of a low thermal gradient and heterogeneous nucleation induced by coaxial feeding is critical to refining the microstructure of materials processed via coaxial WLDED. Through the multi-beam laser WLDED of the coaxial processing head, the tensile properties can be effectively improved.

Meanwhile, Kelbassa et al. [99] assessed the potential of this novel wire-based laser material deposition approach. They constructed material samples of IN718 and titanium Ti6Al4V using coaxial wire feeding laser deposition and obtained samples with extremely low porosity and good metallurgical bonding layers.

Coaxial wire feeding has certain limitations: the wire feeding path is concentrated at the center of the arc. When multiple wires are coordinated, it is easy to cause uneven melting of the wire material due to space limitations, especially for high-melting-point or dissimilar materials, local poor fusion may occur. The diameter and hardness of the wire need to match the size of the coaxial channel, and the material's adaptability is limited. Moreover, the heat input is concentrated in the axial area, which is prone to cause local heat accumulation. It is difficult to control the dimensional accuracy of thin-walled or precision components, and their flexibility in regulating complex gradient components is not as good as that of multi-path non-coaxial wire feeding.

3.2.2. Side-Axis Wire Feeding

The multi-material side-axis wire feeding technology employs multiple independent wire feeding devices arranged in parallel with the welding torch. By controlling the conveying paths and sequences of different metal wires at different times or synchronously, it achieves alternating or combined deposition of multiple materials during the arc deposition process. Its core features are a simple structure, low cost, and easy expansion of material quantity. By adjusting the wire feeding position, timing, and process parameters, dissimilar materials can be directionally deposited in specific areas of the component to form macroscopic material zoning or layered composite structures [97]. Sadhya et al. [88] proposed a novel side-axis wire feeding method. By using three independent wire feeders, they designed a three-wire parallel wire feeding mechanism installed at 120° intervals along the tungsten inert gas welding (TIG) welding torch, as shown in Figure 17. Through the combined effect of front and rear wire feeding, the dependence of the wire-feeding direction on the deposited weld bead is reduced, and it has good arc stability.

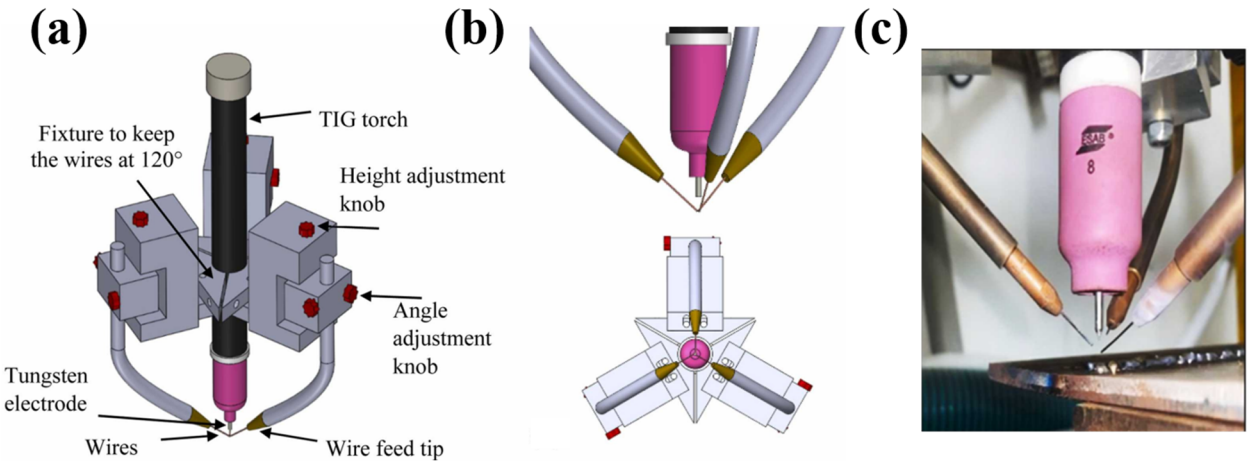


Figure 17. Three-wire feed mechanism diagram: (**a**) Adjustment knob; (**b**) Three torches; (**c**) Three-line feeding mechanism [88].

Zhang et al. [100] investigated the paraxial wire feeding mechanism of plasma arc and TIG arc-based WAAM processes. They use two separate wire feeders to manufacture titanium and aluminum alloys, and reported that favorable dilution could be attained, and furthermore, the feeding rate of the components could be controlled in accordance with the design specifications. Similarly, Liu et al. [101] integrated three deposition welding torches using two metal inert gas welding (MIG) power supplies and indirect connections. The

welding torch adopts two different arrangements, namely the central arrangement and the symmetrical arrangement of the welding wire, that is, the arrangement of the welding wire relative to the welding direction. They found that the deposition rate had been significantly improved due to the increase in melting efficiency. They also reported that by choosing the polarity of the main wire and the side wire feeding gun as the anode and cathode, respectively, two indirect arcs can be superimposed into one indirect arc, thereby achieving high energy concentration and improving the overall welding melting efficiency.

The side-axis wire feeding has obvious advantages over coaxial wire feeding: multiple wires can be arranged through non-central paths, avoiding spatial limitations, achieving independent melting, and precise control of different wire materials. It is particularly suitable for the co-deposition of dissimilar materials or high-melting-point metals, reducing poor fusion. It has a wider range of material adaptability, lower requirements for the diameter and hardness of the wire, and is convenient for the flexible replacement of wire types. The heat input distribution is more uniform, which can reduce local heat accumulation, lower the risk of deformation of thin-walled components, and offer higher flexibility in the regulation of complex gradient components and interface quality control, making it suitable for multi-material composites and high-precision forming requirements.

4. Influencing Factors and Regulation Methods

The core challenges of MMAM are concentrated in two major dimensions: interface quality defects and process control instability. In terms of interface quality, insufficient bonding strength and cracks are caused by the superposition of interface shear stress and phase transformation stress resulting from the mismatch of the material's coefficient of thermal expansion. The brittleness of IMCs overgrows under high heat input, becoming a crack propagation channel. The oxide film hinders the wetting of liquid metal, resulting in incomplete fusion defects. Pores and inclusions form stress concentration sources due to disturbance by protective gas, evaporation of elements, or cross-contamination during wire feeding. At the process control level, uneven heat input leads to distortion of the temperature field in the molten pool due to the difference in thermal conductivity of heterogeneous materials, inducing local overheating and remelting or incomplete fusion. Uneven mixing of the molten pool is manifested as high-viscosity materials inhibiting convection, interlayer temperature waves causing abnormal grain coarsening, and repeated dissolution-precipitation thickening of IMCs, which seriously weakens the interlayer consistency [36,54,102,103]. In response to the above issues, scholars have, respectively, regulated the microstructure from the initial transition layer to the addition of an external field during the process, and finally to the post-treatment to regulate the microstructure and mechanical properties.

4.1. Influencing Factors

4.1.1. Deposition Sequence

In multi-material WAAM, the deposition paths, stacking sequences, and switching timing of different materials precisely control the positions, volumes, and interconnection methods of each material region. When the deposition sequence leads to the encounter of two materials in the molten pool, the dynamic mixing behavior of the molten pool determines the dilution rate, component distribution, phase composition, and possible IMCs formation in the interface region [104]. The materials deposited first will be repeatedly subjected to the thermal cycle of the subsequent deposited layers, and their heat-affected zones may be remelted multiple times or undergo complex heat treatment processes. If the two materials show significant differences in chemical compatibility and thermophysical properties, the selection of the deposition sequence becomes particularly crucial. In addition, the deposition sequence determines the microstructure evolution path of the component,

which in turn affects the overall and local performance. The materials deposited later will impose complex thermal cycles on the materials deposited earlier, which will affect the grain size, phase transformation behavior, precipitated phase state, and dislocation density of the materials deposited earlier [105,106].

Tian et al. [107] prepared the components of Al-6.25Cu/Ti6Al4V dissimilar alloys by CMT technology, and found that the additively manufactured components could be obtained regardless of the deposition sequence, as shown in Figure 18a. When the titanium alloy was deposited initially, a brazed joint formed between the titanium and aluminum alloys, with $TiAl_3$ IMCs generated in the reaction layer, as shown in Figure 18b. In addition, fusion bonding occurs, and $TiAl_3$, TiAl, and Ti_3Al are formed in the thicker reaction layer as shown in Figure 18c. The average UTS of the samples fabricated by depositing Ti alloy and Al alloy first were measured to be 108 MPa and 30 MPa, respectively. All specimens exhibited brittle fracture behavior. The results indicate that prior deposition of Ti alloy is more suitable for CMT-WAAM of Ti/Al dissimilar alloy combinations.

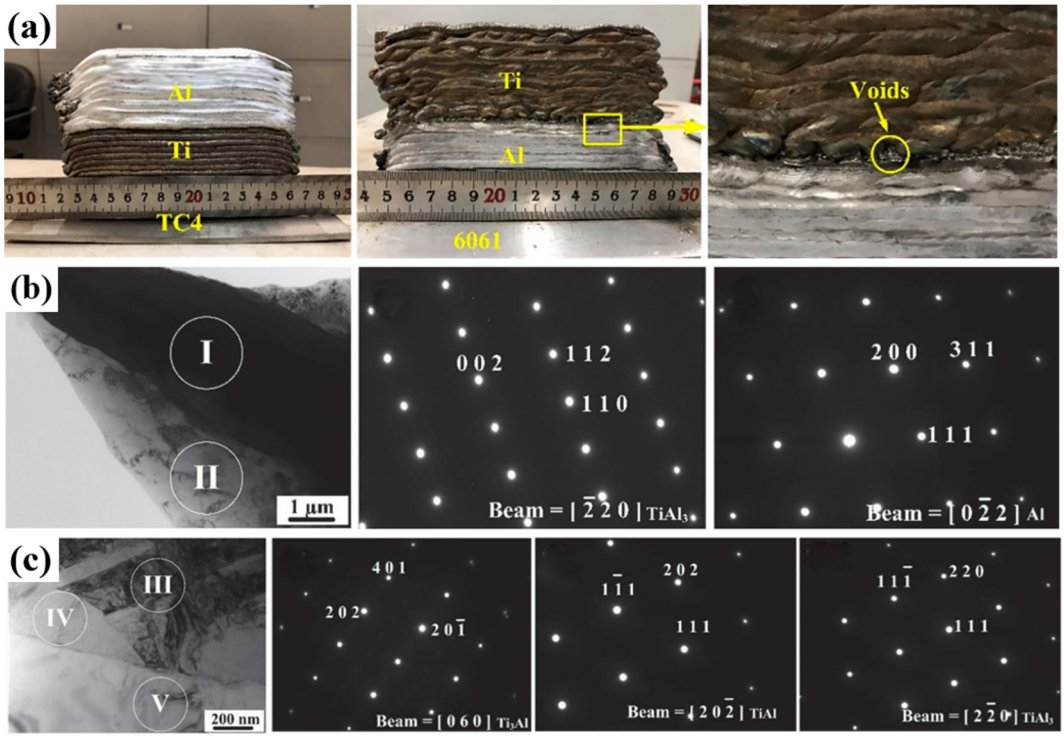


Figure 18. Microstructure analysis of (**a**) Components of sample 1 and sample 2; (**b**) Transmission electron microscopy analysis of sample 1; (**c**) Transmission electron microscopy analysis of sample 2 [107].

Ozaner et al. [108] investigated the effects of deposition sequence and cooling rate on the microstructure and wear properties of SS309L-IN625 alloy structure prepared by WAAM process. Experimental results indicated that the deposition sequence exerted a critical influence on the microstructure and mechanical properties of bimetallic components. Specifically, when IN625 was deposited onto SS309L, the wear resistance was enhanced by 22% compared to the reverse deposition sequence, attributed to the mitigation of thermal exposure and refinement of the interfacial microstructure. The interface region exhibited a mixed adhesive-abrasive wear mechanism, with microhardness values ranging from 285 HV to 340 HV, as influenced by residual stress and interfacial diffusion.

Kennedy et al. [109] reported that when fabricating customized components from two high-performance dissimilar titanium alloys via the WAAM process. The nature of the chemical mixture and microstructure gradient occurring at the interface transition reveals that the α microstructure gradient observed during the transition from Ti6Al4V

to Ti-5Al-5V-5Mo-3Cr is more sudden compared to when the two alloys were deposited in reverse order. In the former scenario, the transformation behavior of Ti6Al4V under the thermal conditions of WAAM exhibits higher sensitivity to β-stabilizing elements in comparison to the scenario where Ti-5Al-5V-5Mo-3Cr undergoes dilution by Ti6Al4V.

In similar work, Chang et al. [110] fabricated two dissimilar alloy components, 2319 and 5B06, using WAAM technology, as shown in Figure 19. When 5B06 alloy is clad onto a 2319 alloy substrate, the elevated thermal input during the deposition process induces significant remelting of the pre-deposited 2319 layer in the interfacial region. This results in an extended cooling duration, which promotes sufficient interdiffusion of alloying elements across the interface. When 2319 alloy is deposited onto 5B06 alloy, the scenario is the exact inverse.

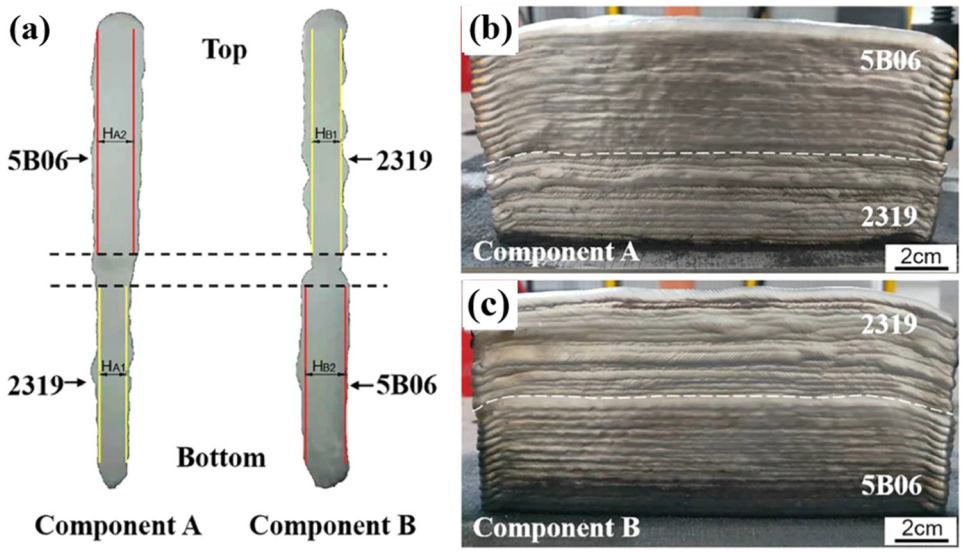


Figure 19. (**a**) Cross-sectional view; (**b**) Single wall of component A; (**c**) Single wall of component B [110].

Figure 20 presents the EBSD maps in the vicinity of the interfacial layers of the two components. Owing to extensive remelting and slow cooling, the interface of Component A lacks distinctness, with no distinct clustered columnar grains observed, as illustrated in Figure 20c. At the bottom region of Figure 20a, fine grains have formed as a result of remelting. In contrast, the interface of Component B in Figure 20d exhibits a distinct columnar grain region. Regarding mechanical properties, the mechanical performances of samples corresponding to Components A, B, and two single-alloy components—all fabricated under identical process parameters—are presented in Figure 20e,f. The results indicate that, compared with Component B, the tensile strength, yield strength (YS), and elongation of Component A are increased by 83.9 MPa, 31.2 MPa, and 4.3%, respectively. The maximum tensile strength and yield strength of Component A are 292.7 MPa and 141.3 MPa, respectively, which are only marginally lower than those of the 2319 base material. It is also noteworthy that the 2319 sample exhibits lower strength than the 5B06 sample. Thus, fracture is likely to initiate on the weaker side. However, regardless of the deposition sequence, the fracture position of the tensile specimen is always approximately 2 mm above the interface layer.

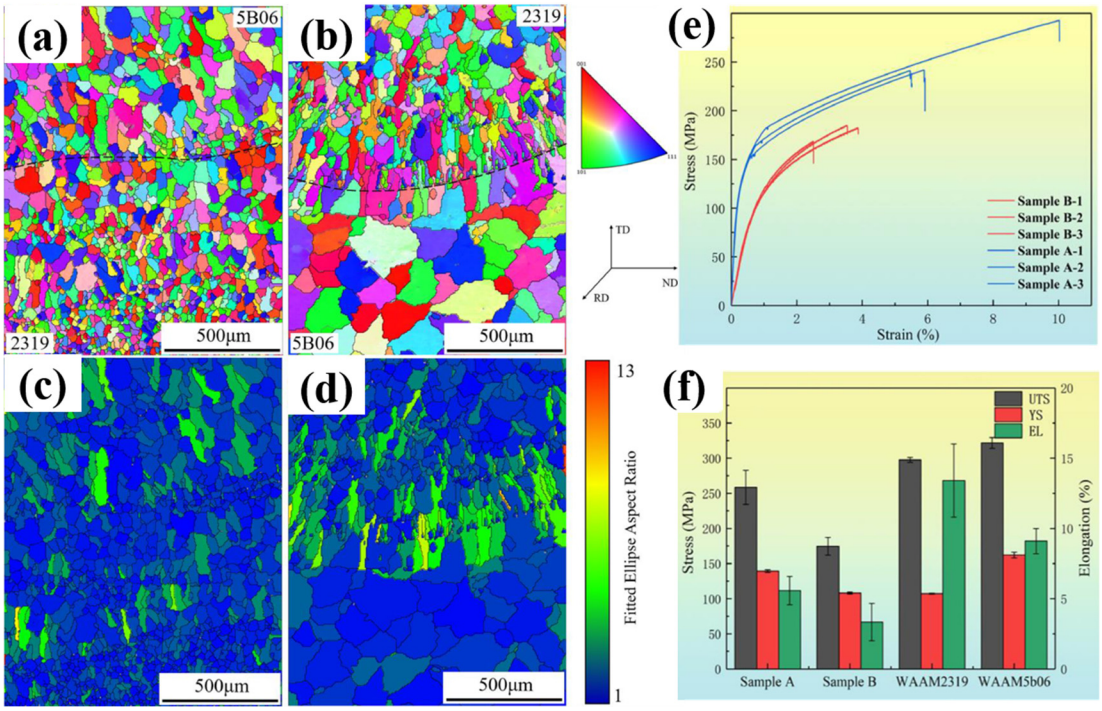


Figure 20. Inverse Pole Figure color maps and fitted ellipse aspect ratios of (**a**,**c**) Component A and (**b**,**d**) Component B; (**e**,**f**) Comparison of mechanical properties of aluminum alloys with different compositions [110].

4.1.2. Deposition Current

The current magnitude serves as a core energy input parameter in MMAM. Its regulation directly affects the thermodynamic behavior of the molten pool, the interaction of heterogeneous materials, the metallurgical quality at the interface, and the stability of the overall performance of the component. Especially in multi-material systems, due to the differences in thermal, physical, and electromagnetic properties of different components, more complex coupling effects are manifested [111].

The current first dominates the size, fluidity, and stability of the molten pool by changing the intensity of heat input, and then determines the material fusion and mixing behavior [103]. Especially when multiple materials are deposited, an increase in current significantly enhances the arc thermal power and droplet transition energy, forming a larger and deeper molten pool, prolonging the existence time of the molten pool, and strengthening internal convection. Although this is beneficial for enhancing the interfacial diffusion and metallurgical bonding of heterogeneous materials, excessively high current may lead to excessive evaporation of low-melting-point components or insufficient melting of high-melting-point components. Especially when the deposition path crosses the material switching boundary, sudden current changes or improper adaptation can cause melt pool collapse, element burning, or incomplete fusion defects. The increase in current intensifies the turbulence of the molten pool, promotes the forced convective mixing of heterogeneous elements, and expands the width of the interfacial dilution zone. However, it may also induce excessive generation of harmful IMCs [21,112,113].

In addition, an increase in current directly raises the heat input of a single layer, which will also lead to a more intense temperature gradient and cooling contraction. When the thermal expansion coefficients of adjacent materials differ significantly, the local high heat input caused by high current will intensify the mismatch strain, generating shear stress concentration near the interface and inducing microcracks or even interlayer delamination [114]. Tian et al. [115] at constant process parameters, and the wall of Ti

(Ti6Al4V)/Al(Al-6.24Cu) dissimilar alloy was fabricated using two different modes of CMT at a gas flow rate of 20 L/min. Due to the formation of cracks at the interface layer in the CMT-pulse mode, the average UTS and hardness value (108 MPa) at the reaction layer in the direct CMT mode are higher than those in the CMT-pulse mode (24 MPa). Kesarwani et al. [116] adopted two deposition directions and three current combinations (115 A/90 A, 120 A/95 A, 125A/100A), as shown in Figure 21.

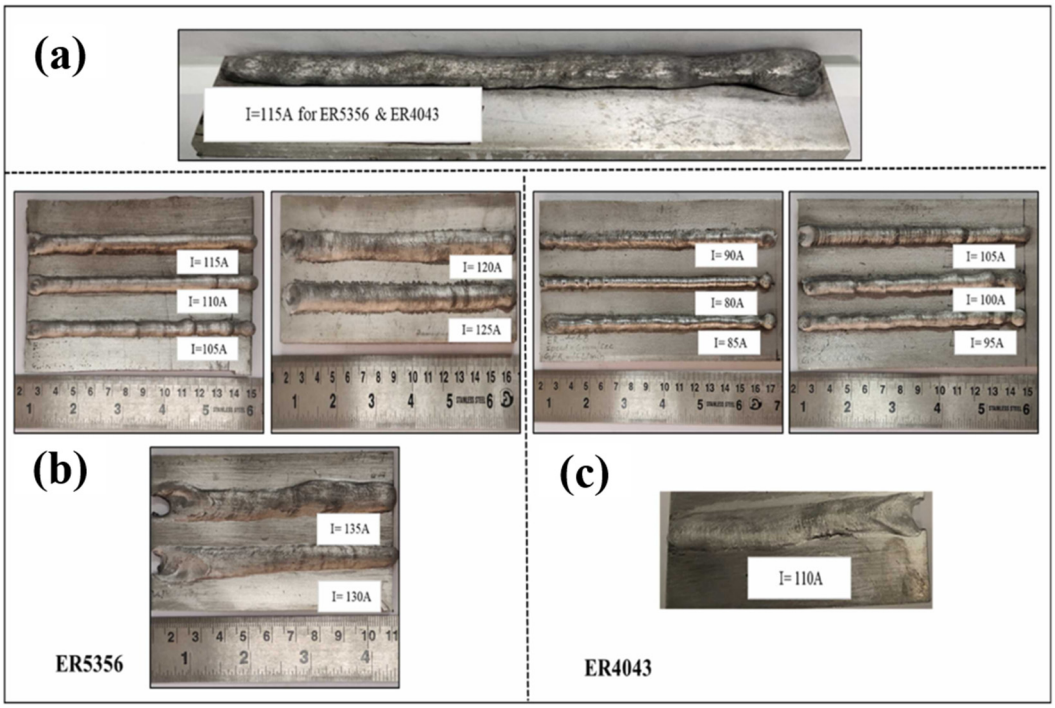


Figure 21. (**a**) Deposition of ER5356 and ER4043 weld beads under the same current condition; (**b**) ER5356 weld beads under different current conditions; (**c**) ER4043 weld beads under different current conditions [116].

Effects of deposition direction and welding current on the microstructure and properties of ER5356/ER4043 alloy inner walls via the CMT arc additive manufacturing process were investigated. The experimental results show that under the current combination of 115A/90A, the performance of bidirectional wall construction is better. Furthermore, under the 115A/90A current combination, the hardness in the transverse direction is enhanced; additionally, the hardness of the bidirectionally deposited wall is also higher than that of the unidirectionally deposited wall. Moreover, bidirectional deposition induces lower levels of residual stress compared to unidirectional deposition.

4.1.3. Wire Feeding Speed

In WAAM multi-material manufacturing, the WFS is one of the core process parameters, which directly affects the material fusion quality, deposition efficiency, and interface performance of dissimilar materials. Song et al. [117] proposed a method for preparing Ni-Ti alloys by alternating current filling with double welding wires. During this process, the Ni-filled wire and Ti-filled wire were connected to the alternating current (AC). The macroscopic morphology, phase evolution, and mechanical properties of the additively manufactured samples at different WFS ratios were investigated. It was found that during the deposition process, the sedimentary layer was always uniform. As the feed rate of the nickel wire decreases, the coarse dendrite structure gradually changes. Columnar and equiaxed grain structures were observed in the additively manufactured deposits under varying WFS ratios. As illustrated in Figure 22, the average microhardness exhibits an

increasing trend with the increase in nickel wire feeding rate. At a Ni/Ti feeding ratio of 1.2, the compressive strength reaches 1664 MPa.

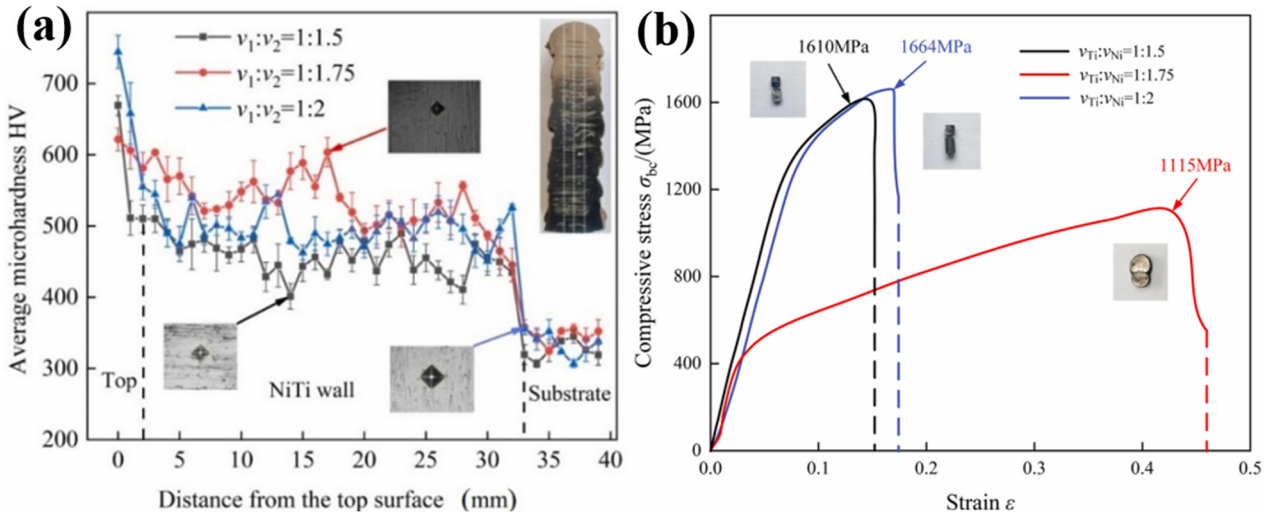


Figure 22. (**a**) Microhardness distribution and measured locations on the as-deposited part; (**b**) Stress–strain curves of the compression test [117].

Xiong et al. [118] reported that a higher WFS would increase the surface roughness of WAAM stainless steel components. Figure 23 shows the surface appearance of the components deposited at different WFS. In Figure 23a,b, the boundaries of adjacent layers of the component are distinguishable. The surface morphology presented in Figure 23b is nearly identical to that in Figure 23a. When the WFS was increased to 4.92 m/min, as illustrated in Figure 23c, a mixed-layer phenomenon was observed in a specific region, indicating that the molten pool was unstable during the multi-layer deposition process. Furthermore, the correlation between surface roughness and WFS is presented in Figure 23d. The surface roughness increases with increasing WFS.

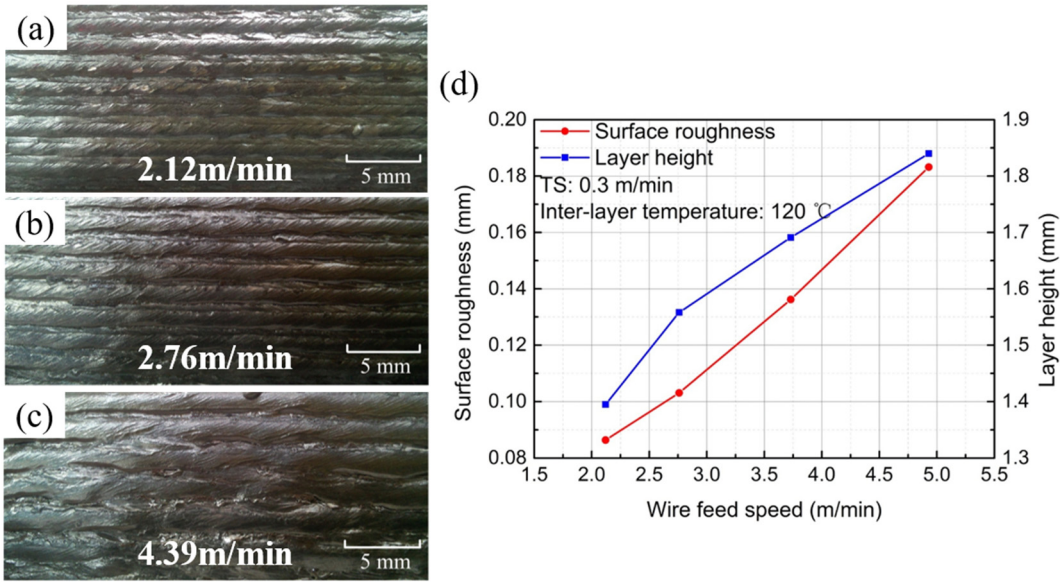


Figure 23. (**a**–**c**) Surface morphology of thin-walled components deposited under different WFS conditions; (**d**) Effect of WFS on the surface roughness of thin-walled components [118].

In summary, the WFS is a key factor in the WAAM process. By controlling the material input rate, it profoundly and extensively influences the behavior of the molten pool, the

geometric morphology of the deposited layer, the thermal cycling process, the evolution of mechanical properties, microstructure, production efficiency, and material utilization rate. Precise control and optimization of the WFS and its matching with other process parameters constitute the core prerequisite for realizing high-quality, high-efficiency, and high-performance WAAM.

Current research on the interface of heterogeneous materials has largely fallen into the trap of superficial representation. Many studies are content with presenting an interface morphology diagram without macroscopic cracks and a set of relatively good mechanical property data, and thus are eager to declare success. However, this is far from touching upon the core of the issue. We still lack a profound understanding of the nanoscale phase formation sequence at the interface, the kinetics of element mutual diffusion, the three-dimensional distribution of residual stress, and the exact mechanism of crack initiation. This lack of research depth has made it difficult for us to establish reliable process-organization-performance prediction models up to now, resulting in process development still largely relying on the trial-and-error method. This is an obstacle that must be overcome for this field to mature.

4.2. Regulation Methods

4.2.1. Transition Layer

In WAAM multi-material forming, the intermediate transition layer can significantly improve the bonding quality at the interface of dissimilar materials by choosing materials with good compatibility with both matrix materials or designing gradient components. On the one hand, the transition layer can promote atomic diffusion and metallurgical bonding between different materials, reduce the formation of brittle phases at the interface or defects such as pores and cracks caused by excessive differences in chemical composition, and thereby enhance the interfacial bonding strength. On the other hand, it can alleviate the thermal stress concentration caused by poor thermal matching of materials during the deposition process by gradually adjusting the thermal physical properties, thereby reducing the risk of component deformation or cracking. Meanwhile, the microstructure of the transition layer may also improve the overall mechanical properties of multi-material components by regulating the solidification behavior and coordinating the mechanical property differences in adjacent materials. In addition, in functional multi-material forming, the transition layer can reduce problems such as galvanic corrosion caused by the contact of dissimilar materials, and enhance the comprehensive performance of the component, such as corrosion resistance and wear resistance [36,119,120].

Manohar et al. [121] investigated the mechanical properties and microstructure of SS304L-CuNi-AA2319 polymetallic structures fabricated via the WAAM process. A Cu-Ni intermediate layer was employed to join SS304L stainless steel and AA2319 aluminum alloy. This Cu-Ni intermediate layer mitigates the formation of harmful IMCs, thus enabling the formation of a defect-free polymetallic structure with improved bonding performance. Tensile testing results indicated that the polymetallic structure exhibited a tensile strength of 185 MPa, ranking the highest among analogous aluminum-steel composite structures fabricated via WAAM. The influence of interlayer and interface microstructure on mechanical properties has been highlighted. In addition, Zhang et al. [122] fabricated a multi-layer IN718 intermediate layer composite structure of QCr0.8 HSHC Cu alloy/S06 stainless steel. The results showed that there were no obvious exposed defects at the interface between QCR0.8/IN718 and IN718/S06, and the combination was good. The interface of QCr0.8/IN718 presents the characteristics of a combination of columnar crystals and equiaxed crystals, as shown in Figure 24a. It can be seen from Figure 24b that the area near

the interface shows obvious strain accumulation. This is due to repeated thermal cycles. This also confirms the occurrence of metallurgical bonding.

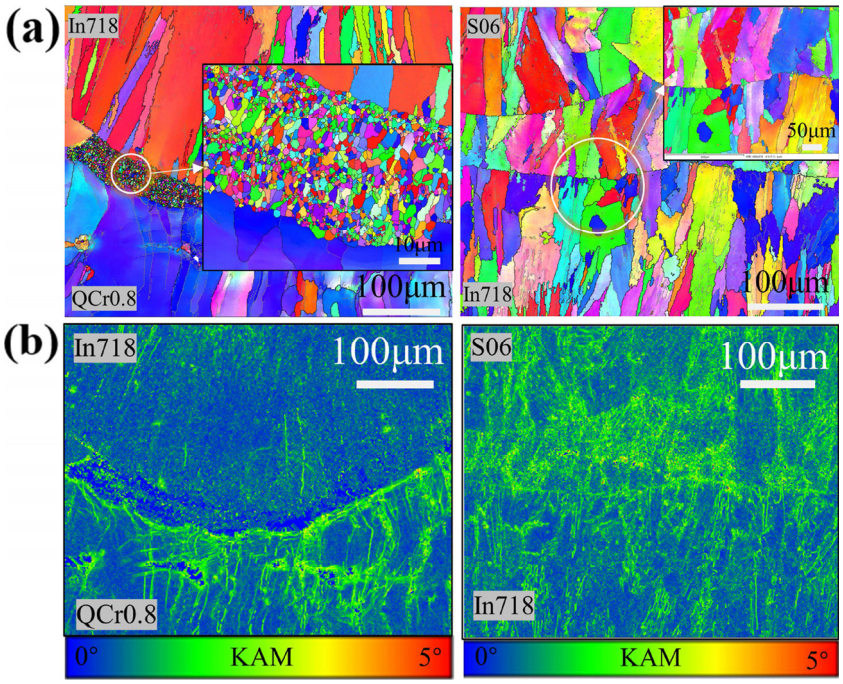


Figure 24. EBSD results of the interface. (**a**) IPF map; (**b**) KAM map [122].

Paul et al. [123] used copper as the intermediate layer and deposited the SS-Al transition wall by arc DED technology. At the SS304L-Cu interface, the formation of continuous Fe-Cu and Fe-Si solid solutions was observed, rather than that of IMCs. In contrast, three distinct types of IMCs were formed at the Al-Cu interface, namely Al-Cu binary IMCs, Al_2Cu, and Al_4Cu_9. Additionally, the study observed that Al and Al_2Cu formed a lamellar structure within the supersaturated solid solution. Mechanical testing results revealed that the Al-Cu interface undergoes failure without significant plastic deformation, indicating the brittle nature of the interface.

Xu et al. [124] proposed a method of adding a Nb intermediate layer through tungsten inert gas welding and applied it to the WAAM technology based on CMT technology to prepare crack-free Ti6Al4V/Al-6.21Cu composite structures.

It was observed that the introduction of a Nb intermediate layer between Ti6Al4V and Al-6.21Cu could effectively impede the diffusion of Ti and Al elements, thereby inhibiting the formation of brittle Ti-Al IMCs, as illustrated in Figure 25.

A reaction layer consisting of Al_3Nb was formed at the Al-6.21Cu/Nb interface, as illustrated in Figure 26. The average UTS of the Ti6Al4V/Al-6.21Cu BS with and without the Nb interlayer were determined to be 120.9 MPa and 94.5 MPa, respectively. This indicating that the introduction of the Nb intermediate layer can significantly enhance the mechanical properties of the aforementioned BS.

Inspired by biological heterogeneous lamellar structures, Jiang et al. [125] incorporated a Nb intermediate layer into the WAAM of Ni-Ti alloy and TC4 alloy to inhibit the intermixing of alloying elements from both sides. Consequently, the presence of Ti_2Ni IMCs was significantly diminished, and the mechanical properties of NiTi-Nb-TC4 alloy components were enhanced.

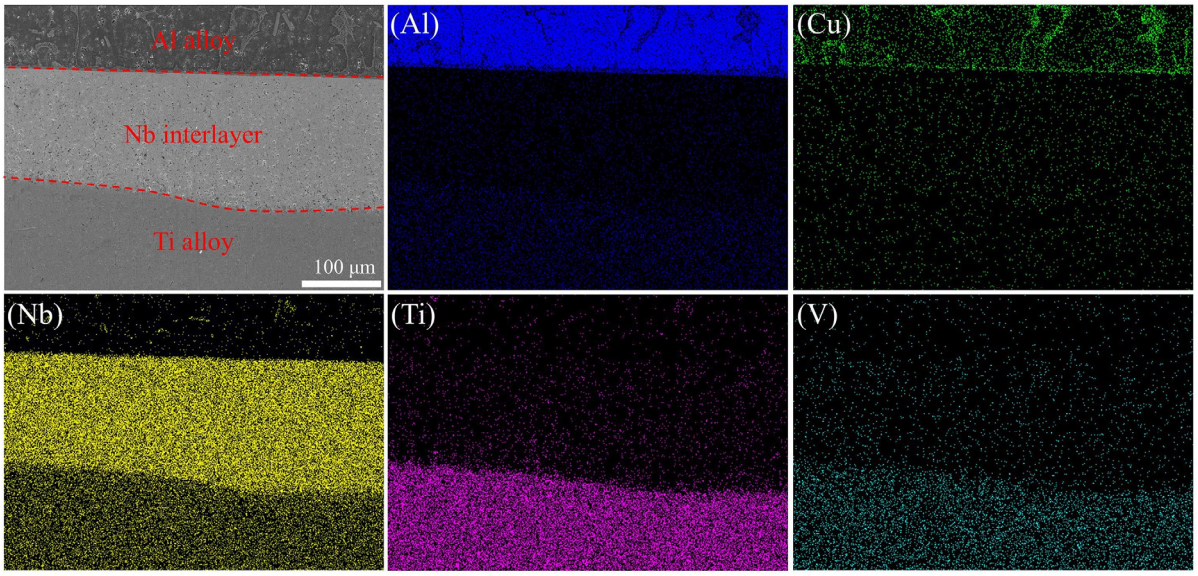


Figure 25. EDS map of the interface of the BS with Nb interlayer [124].

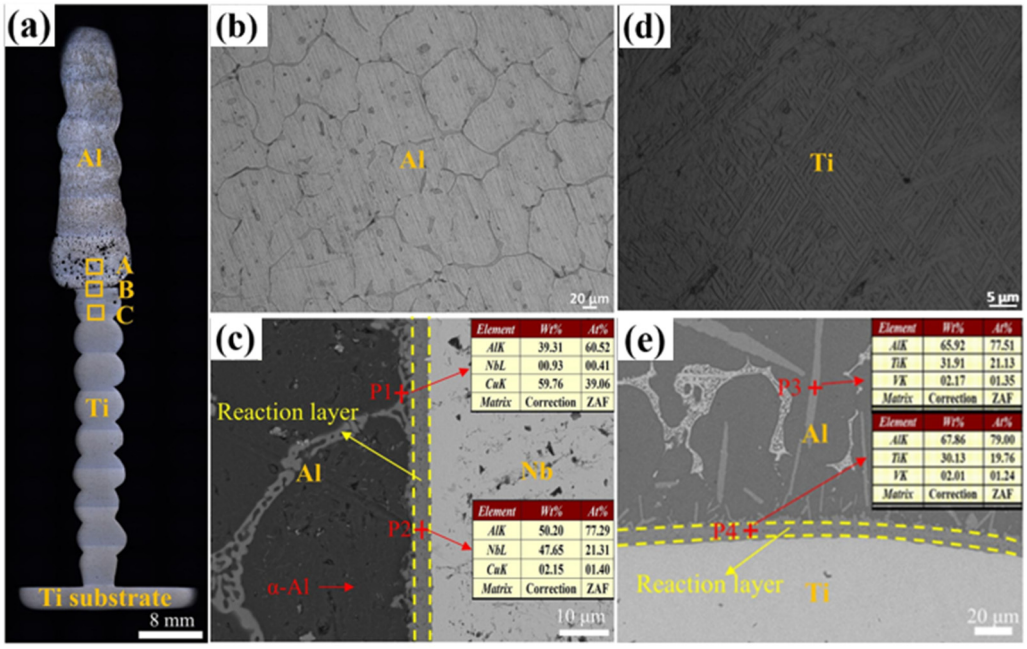


Figure 26. (**a**) Cross-sectional view of the BS with a Nb interlayer; (**b**–**d**) Microstructures of Zones A–C; (**e**) Interface microstructure of the BS without a Nb interlayer [124].

The study revealed that the compressive strength of the WAAM-fabricated NiTi-Nb-TC4 alloy was determined to be 1549.0 ± 17 MPa, with a fracture strain of 22.3 ± 8% in the direction perpendicular to the interface. Its compressive strength corresponded to 63.6% and 94.5% of those of the WAAM-fabricated NiTi alloy and TC4 alloy, respectively, as illustrated in Figure 27a,b. The introduction of the Nb intermediate layer enhanced the tensile strength in the direction perpendicular to the interface to 460.7 ± 16 MPa. The tensile strengths perpendicular to the interface of the WAAM NiTi-Nb-TC4 composite specimens can reach 72.0% and 60.0% of the WAAM NiTi alloy and TC4 alloy, respectively, as shown in Figure 27c,d.

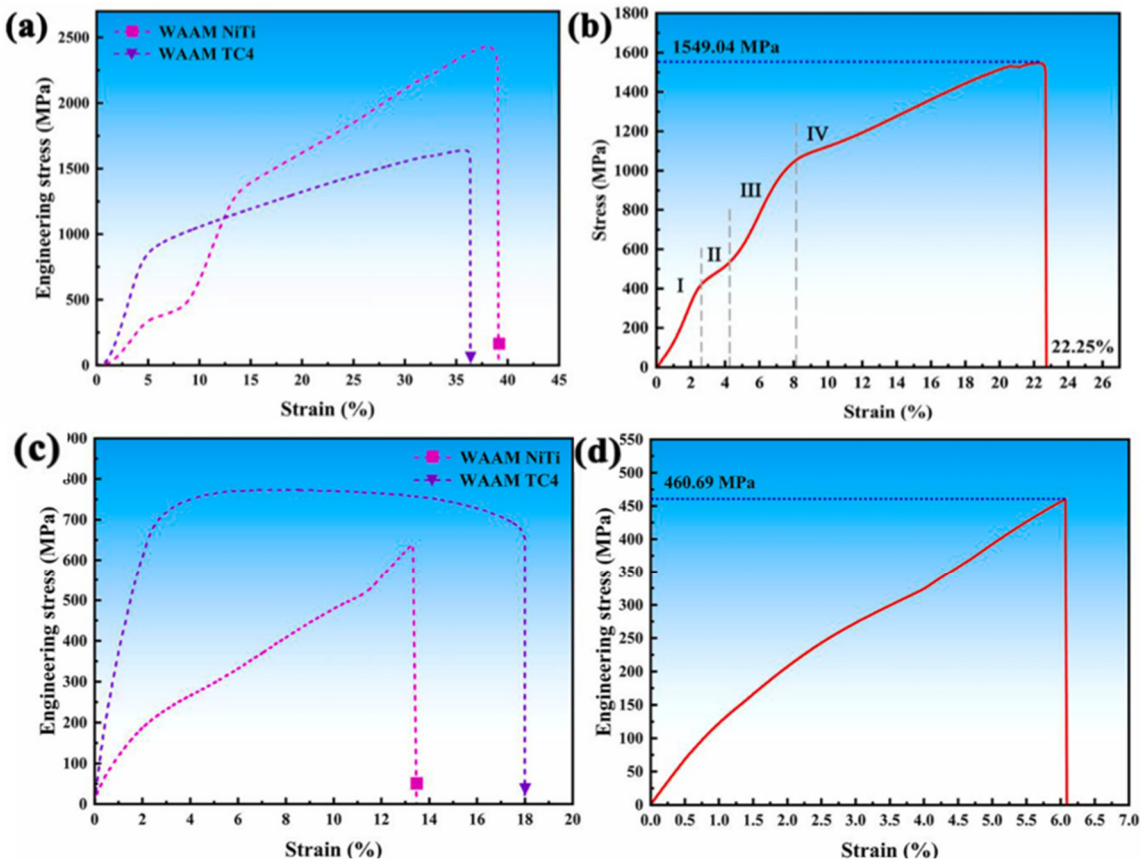


Figure 27. Compressive properties of (**a**) WAAM-fabricated NiTi and TC4 samples; (**b**) WAAM-fabricated NiTi–Nb-TC4 heterogeneous alloy composite samples; (**c**) Tensile properties of WAAM-fabricated NiTi and TC4 samples; (**d**) WAAM-fabricated NiTi–Nb-TC4 heterogeneous alloy composite samples [125].

It is worth mentioning here that in the face of the challenges of interface control, many of the schemes proposed in the current literature demonstrate academic ingenuity, but the feasibility of their industrial transformation deserves in-depth investigation.

4.2.2. External Field

The application of an external field has a synergistic optimization effect on the multi-material deposition of WAAM. It precisely regulates the kinetics of the molten pool through non-contact energy input, refines grains and alleviates thermal stress, efficiently breaks dendrites and oxide films, and accelerates element diffusion. Comprehensively enhancing the fatigue life, dimensional accuracy, and service reliability of multi-material components is the key technical path to achieving the manufacturing of high-performance gradient functional materials [126].

The external ultrasonic field is widely applied in the WAAM multi-material field. In the WAAM multi-material deposition, the ultrasonic field generates unique effects through high-frequency mechanical vibration. Its cavitation effect instantly forms high-pressure micro-jets in the molten pool, efficiently breaking dendrites, refining grains to the micrometer level, and accelerating element diffusion, significantly improving the composition uniformity of heterogeneous materials. The strong acoustic flow effect enhances the convection of the melt, promotes the upward floating of bubbles and impurities, and reduces the porosity. Cavitation energy can break the oxide film and activate interfacial reactions, inhibit the continuous growth of brittle IMCs, and enhance the interfacial bonding strength. High-frequency vibration can also relieve lattice distortion, mitigate residual stress, and

diminish the risk of cracking arising from differences in thermal expansion coefficients between multiple materials. In addition, ultrasonic-induced non-uniform nucleation optimizes the microstructure of the transition zone of dissimilar materials, comprehensively enhancing the fatigue life and reliability of components. This technology constitutes a high-precision energy field-assisted technique, and notable progress has been achieved in the domain of MMAM to date [127,128].

Jeong et al. [129] studied the effects of ultrasonic treatment on the microstructure and mechanical properties of functional gradient materials of SS308L and IN718 manufactured by double-wire arc additive manufacturing. The schematic diagram of the principle is shown in Figure 28.

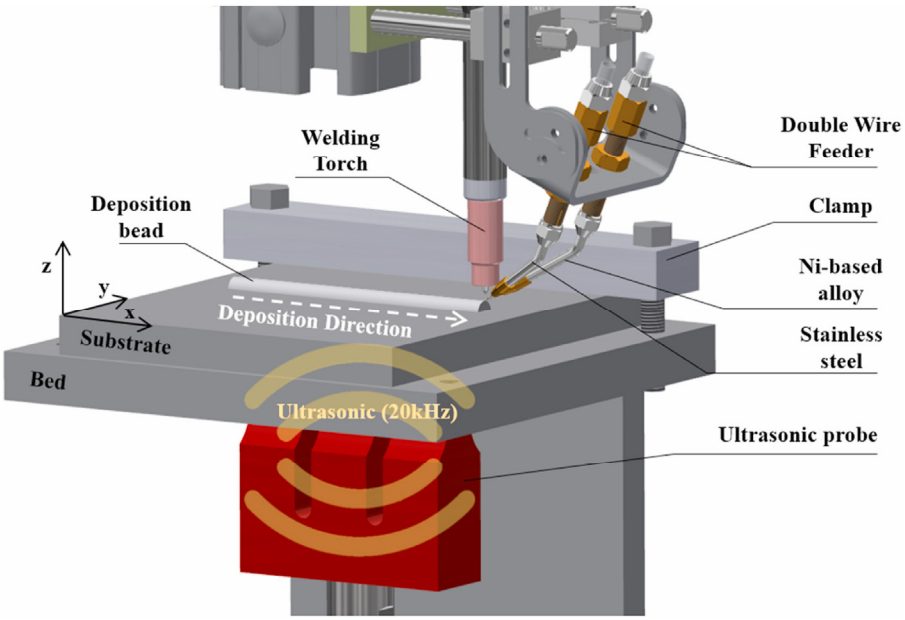


Figure 28. Schematic illustration of the ultrasonic-assisted double WAAM system [129].

It was observed that without ultrasonic treatment (UT), columnar grains with strong texture along the building direction were formed on the cross-section of SS308L. Conversely, after UT, the columnar grains transform into equiaxed grains. Specifically, in the pre-UT sedimentary structure, as shown in Figure 29a–e, columnar dendrites similar to the deposited IN718 alloy were observed in the 67% IN718, 50% IN718, and 33% IN718 regions. After UT, all the above-mentioned regions were composed of equiaxed dendrites, and the uniformity of the microstructure was significantly enhanced, as shown in Figure 29f–j. This refinement of microstructure leads to enhanced mechanical properties, including hardness and tensile strength.

With an increase in the mixing proportion of IN718, the hardness exhibits a gradual increasing trend from the SS308L region of the FGM toward the direction of increasing IN718 mixing proportion. Compared with the same layer without ultrasonic treatment, the hardness increases percentages after ultrasonic treatment, and without ultrasonic treatment were 7%, 9%, 10%, 12% and 15%, respectively, as shown in Figure 30. The enhancement in hardness is attributed to microstructural evolution, solid solution strengthening, and precipitation strengthening, as the contents of Ni, Mo, and Nb increase concomitantly with the elevated IN718 content.

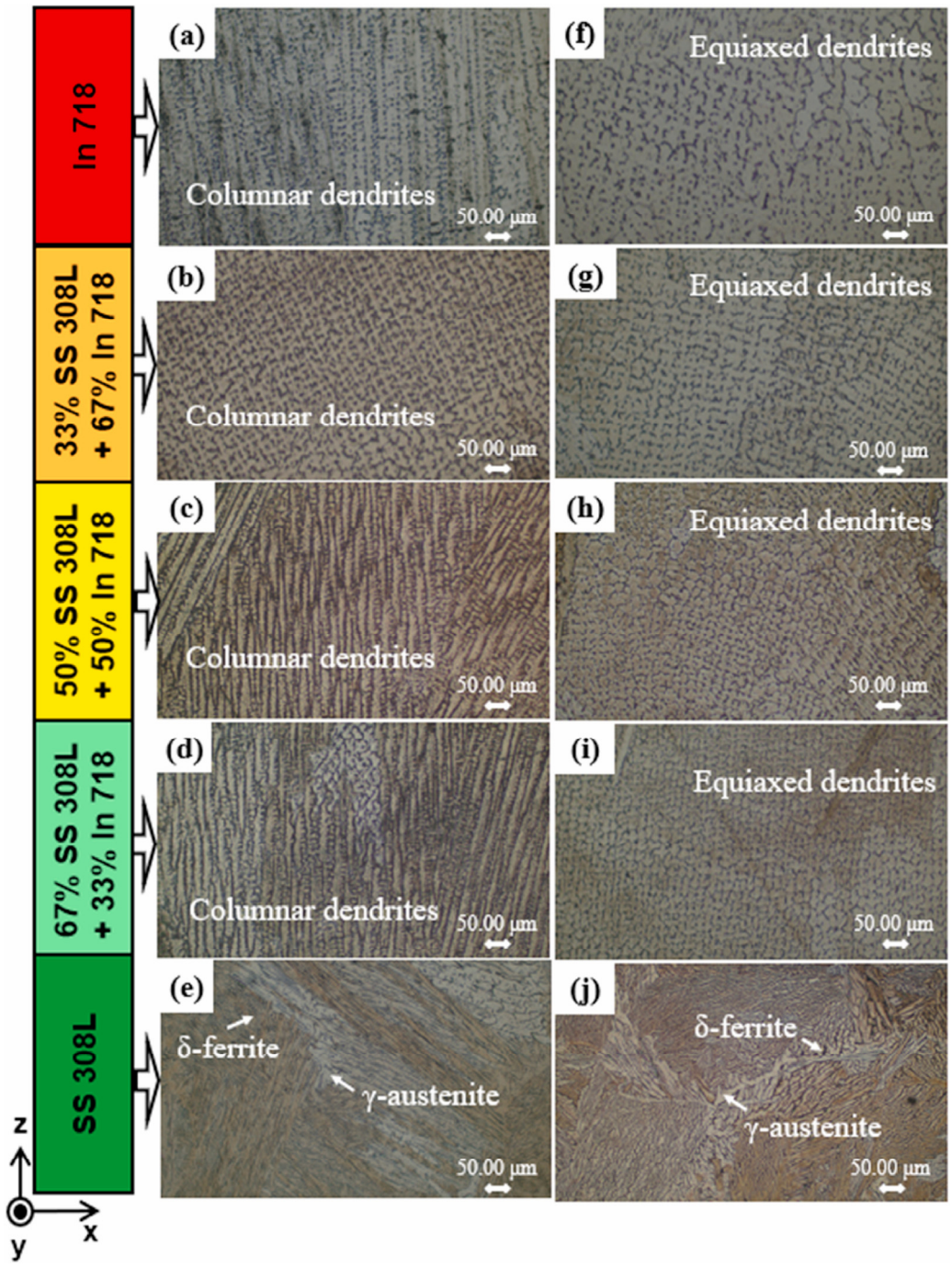


Figure 29. Optical images of the cross-sections of the prepared multi-materials show samples that have not undergone ultrasonic treatment (**a**–**e**) and have undergone ultrasonic treatment (**f**–**j**) under different composition gradients from pure SS to pure IN718 [129].

Xu et al. [130] refined the microstructure of Ti6Al4V/Al-6.21Cu bimetallic components with a Nb intermediate layer-fabricated via CMT-based additive manufacturing—through layer-by-layer ultrasonic peening treatment (UPT), thereby reducing both the number and size of pores within the coarse dendritic structure. Following UPT, the pores within the deposited aluminum layer are compressed and compacted. Under the action of UPT, the grains in the transition layer are significantly compressed and refined. The Nb intermediate layer still functions as a diffusion barrier layer, preventing the formation of harmful Ti-Al IMCs. The average tensile strength of the duplex steel structure with Nb intermediate layer after UPT is 164.5 MPa, which is 36.1% higher than that of the duplex steel structure without

Nb intermediate layer. The fractures of all the specimens occurred in the brittle Al_3Nb reaction layer. Acoustic flow enhances diffusion velocity and high-temperature diffusion, thereby rendering the temperature distribution more homogeneous and mitigating the temperature gradient.

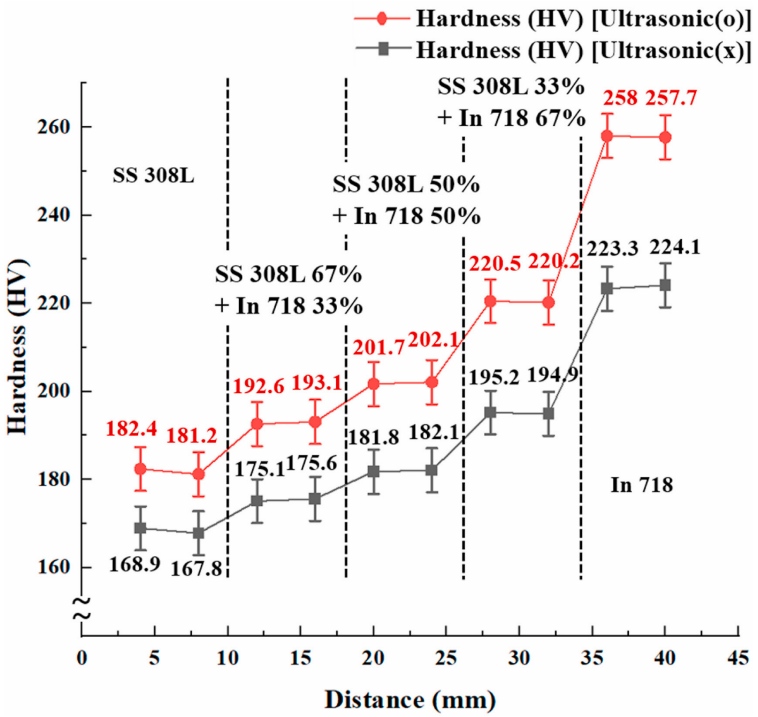


Figure 30. Hardness distribution of FGM as a function of location [129].

In the additive manufacturing process, there is often a problem that the temperature of the molten pool is difficult to control well. Both excessively high and low temperatures can affect the quality of the formed samples. Combining simulations can provide a clearer understanding of the trend of temperature gradient changes. Ji et al. [131] investigated the effect of ultrasonic vibration assistance on the melt pool temperature of WAAM components. As illustrated in Figure 31, the maximum temperature was distributed in the central region and decreased gradually from the center to the edge. This is attributed to the fact that heat dissipation is the slowest in the central region, while it accelerates progressively with increasing proximity to the edge. Additionally, the temperature at the bottom of the melt pool is significantly lower than that at the surface.

This is because the heat transfer coefficient of the metal substrate at the bottom of the melt pool is significantly higher than that of the ambient air above the melt pool surface. With an increase in ultraviolet radiation, the flow velocity of the melt pool increases, which facilitates heat exchange the high-temperature region expands markedly and the temperature distribution becomes more homogeneous. However, the temperature at the bottom of the melt pool does not increase, attributed to two factors: Firstly, the flow velocity in the bottom region is relatively low, leading to reduced heat transfer from the high-temperature region; Secondly, the high heat transfer efficiency of the substrate enables greater heat dissipation to the metal substrate. Eventually, it is found that the acoustic flow promotes the flow rate and high-temperature diffusion, making the temperature distribution more uniform and reducing the temperature gradient.

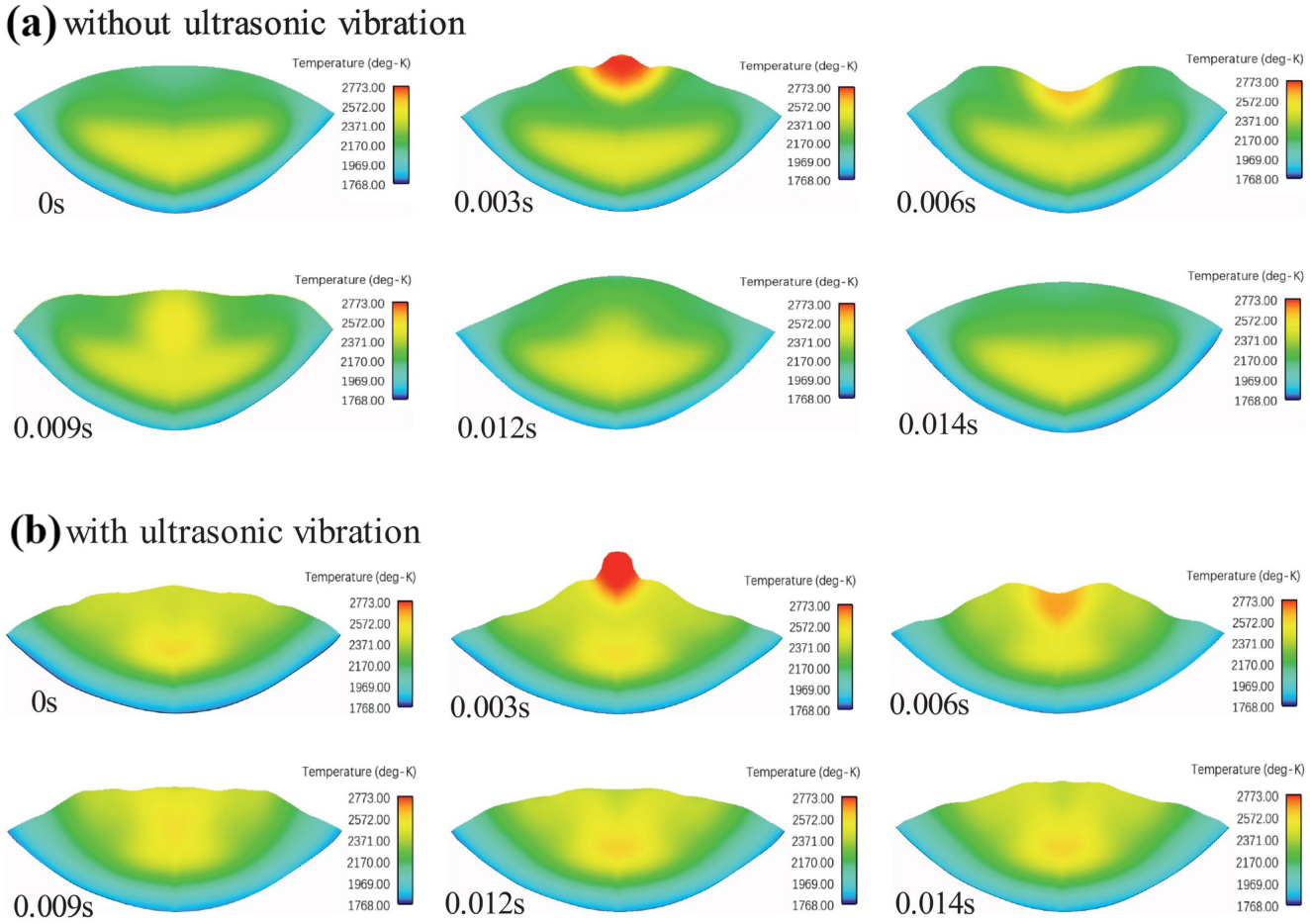


Figure 31. Temperature field in the melt pool at different time points during a single droplet transfer cycle: (**a**) Without ultrasonic vibration assistance; (**b**) With ultrasonic vibration assistance [131].

What needs to be pointed out here is that although external field-assisted technologies such as ultrasound and electromagnetism have shown remarkable effects in laboratories, the reliability, stability, and additional costs and complexity associated with their integration with large industrial robots have been avoided by the vast majority of studies. The academic community urgently needs to collaborate with enterprises to conduct research based on industrialized equipment and cost considerations, rather than merely being content with manufacturing display items in an ideal laboratory environment.

4.2.3. Post-Treatment

Some materials, such as nickel-based superalloys, steels, and others, require post-weld heat treatment (PWHT) to ensure their mechanical properties [132,133]. For single material structures, heat treatment of most materials is well known and follows a specific heat treatment program, as each specific material has its own microstructural response to heat treatment. Additively manufactured materials undergo inherent heat treatment during the manufacturing process, so in some cases, the starting point for PWHT is different, but the basic mechanism of PWHT for additively manufactured materials is comparable to that of conventional materials. In the context of multi-material components, there are differences in interest in the use of different heat treatment methods.

Particular emphasis is placed on four distinct aspects, which are regarded as the key to shaping the final material performance. These aspects include peak temperature, holding time, cooling/heating rate, and number of repetitions. However, considering the inherent variability of the set points between materials, the use of set points in multi-

material components may be beneficial to some materials while causing irreparable damage to others. There are few or even no studies on this topic, but many research topics can emerge here, as shown in Table 3. Furthermore, Yadav et al. presented a method for FGM composed of low-alloy steel and high-alloy steel [134]. Singh et al. [135] on arc additive manufacturing of NiTi-Cu BS, the main focus was on the deposition of nickel-titanium onto copper using WAAM. They conducted an in-depth investigation into the effect of heat treatment and found that the performance was optimal at a heat treatment temperature of 500 °C and a holding time of 12 h, exhibiting a microhardness of 485 HV. The compressive strength can reach 650 MPa.

Also can promote atomic diffusion at the interface of dissimilar materials by precisely regulating the temperature [136], reduce the excessive precipitation of brittle IMCs, and simultaneously release the residual stress accumulated during the deposition process due to the difference in thermal expansion coefficients [137,138]. Zhai et al. [92] deposited the BS of ER70S-6/S355 through WAAM. After quenching heat treatment, the yield strength of the ER70S-6/S355 welded joint fabricated by WAAM was 511.6 MPa, UTS was 740.7 MPa, and the elongation was 20.1%.

In addition, local heat treatment can selectively refine the interfacial grains, inhibit the growth of columnar crystals, and make the interfacial hardness gradient more gentle [139,140].

Kim et al. [141] proposed a BS of IN625 and austenitic SS316L based on WAAM technology. As illustrated in Figure 32, the bimetallic interface is clearly distinguishable in both the as-deposited and heat-treated samples. IN625 has dendrites and columnar structures, while SS316L has columnar austenite grains. Dendrites and sub-grains inoculate δ-ferrite between austenites. At the interface of SS316L, two types of δ-ferrite were found, which were acicular δ-ferrite and skeletal δ-ferrite. Needle-like δ-ferrite is more likely to form at the melting interface than skeleton-like δ-ferrite because heat dissipates from the interface more quickly compared to the interior of the matrix.

Following heat treatment at 970 °C, SS316L undergoes a phase transformation from δ-ferrite to austenite. The δ-ferrite content in H10 and H60 heat-treated steels is reduced by 16–20% compared to that in the as-built samples, which contributes to the enhancement of hardness and mechanical strength. The tensile stress–strain behavior, fracture location, and fracture morphology are presented in Figure 33. The YS of the as-built, H10, and H60 specimens are 322 MPa, 315 MPa, and 300 MPa, respectively. With the prolongation of heat treatment duration, YS exhibited a slight decrease; however, the UTS of the heat-treated specimens were marginally higher than that of the as-built state, with values of 582 MPa, 601 MPa, and 602 MPa, respectively. None of the specimens fractured at the bimetallic interface—all failed on the SS316L side, as shown in Figure 33b. Since SS316L consists of a FCC austenitic phase, the fracture surfaces exhibit ductile fracture features, as presented in Figure 33c.

Table 3. Literature review on the influence of heat treatment on multi-materials fabricated via WAAM.

Material	Heat Treatment Time and Temperature	Effect	Ref.
IN625–SS	970 °C for 10 and 60 min	Increases hardness and strength	[141]
Al Bronze–Steel	650 °C for 6 h	Redirects the grains and homogenizes the microstructure of Al bronze	[55]
Fe_3Ni–FeNi	900 °C held for 1800 s and cooled to 200 °C	Homogenizes the phase volume fraction of materials	[142]
SS–IN625	800 °C for 2 h	Leads to the formation of the IMCs phase and carbides	[143]
IN625–HSLA	1080 °C for 1 h/1080 °C for 2 h	Dissolves the Laves phase	[144]
LCS–SS	800 °C, 950 °C, and 1100 °C for 0.5, 1, and 2 h	Transferring the failure from the LCS side to the SS side increased the elongation.	[145]

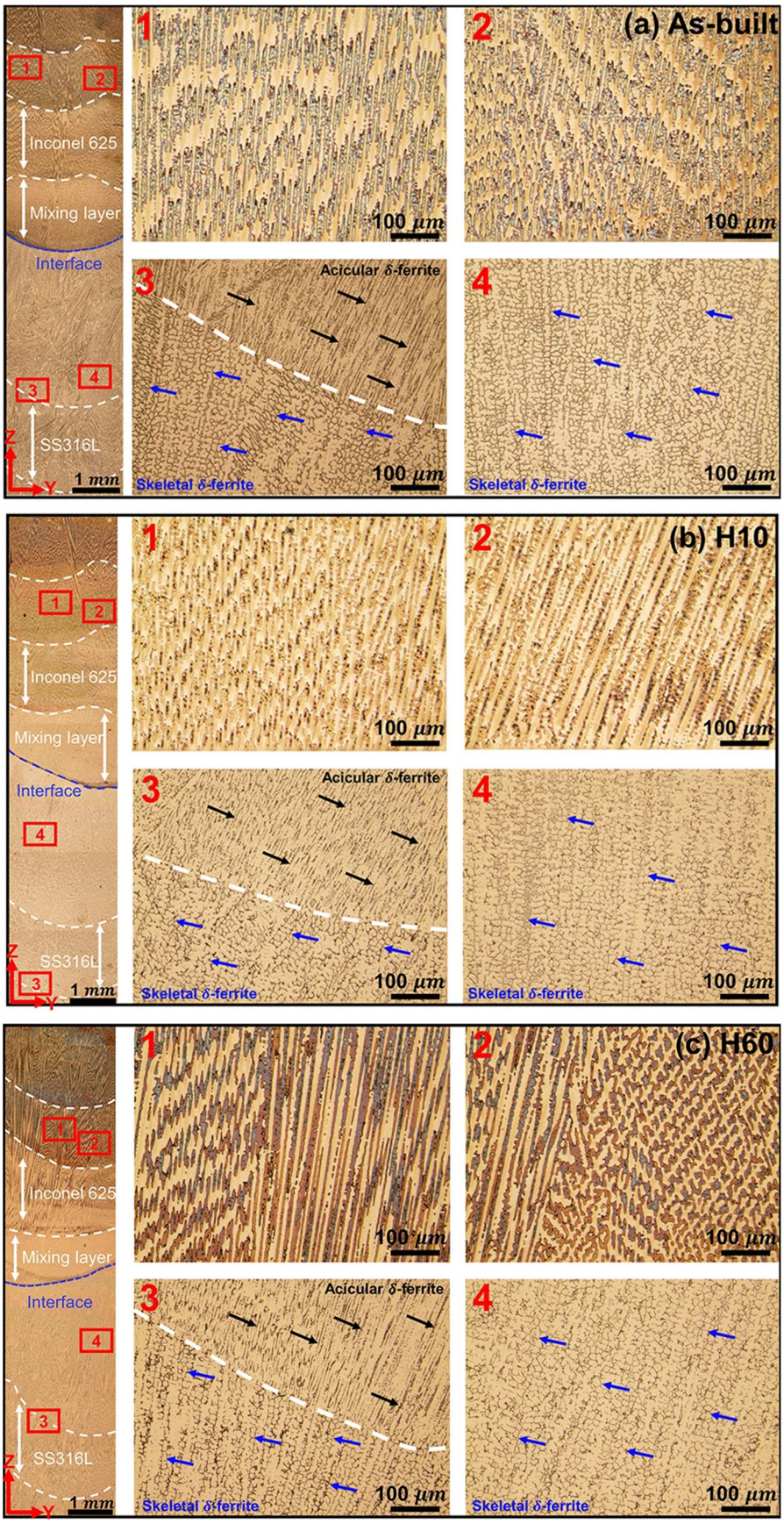


Figure 32. Optical microscopy micrograph depicting the microstructure of (**a**) As-built, (**b**) H10, and (**c**) H60 [141].

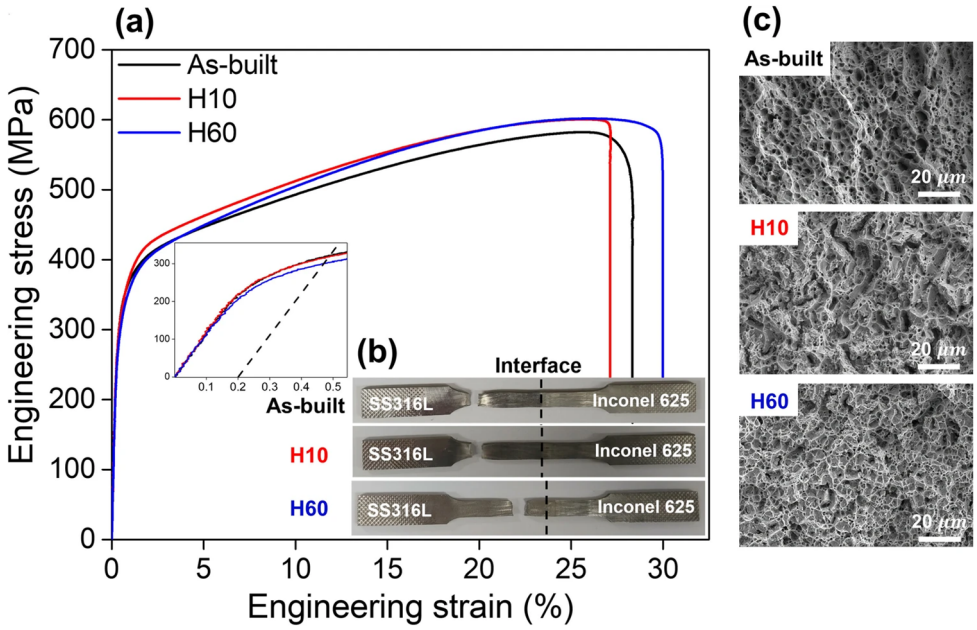


Figure 33. (**a**) Stress–strain curve; (**b**) Fractured specimen after testing; (**c**) Fractography of the BAMS sample [141].

In conclusion, the quality control of multi-material interfaces in additive manufacturing is the core link in achieving precise forming of high-performance multi-material components. The deposition sequence directly determines the interfacial bonding mode and mechanical property differences by influencing the interaction behavior of the material's molten pool, the thermal cycling effect, and the formation mechanism of the reaction phase. Deposition current, as the core parameter of energy input, profoundly affects the width of the interfacial dilution zone, the generation of IMCs, and the distribution of residual stress by regulating the thermodynamic behavior of the molten pool and the element diffusion process. The WFS plays a crucial role in the stability of the molten pool, the deposition efficiency, and the uniformity of the interface composition through the dynamic matching of the material input rate [146,147].

In response to the above-mentioned influencing factors, the transition layer technology effectively inhibits the formation of harmful IMCs and alleviates thermal stress concentration by introducing compatible intermediate phases. The external field auxiliary means achieves the optimization of molten pool dynamics and grain refinement through energy field regulation. The post-treatment process further improves the interfacial metallurgical quality and microstructure state through thermal action and stress release. These regulatory strategies have comprehensively enhanced the interfacial bonding strength, mechanical matching, and defect control capabilities of multiple materials from multiple dimensions, laying an important technological and theoretical foundation for the performance optimization of MMAM components. Mechanical property characterization is essential for quantifying new material systems. In additive manufacturing, it is a key method for comparing AM and conventional components to prove equivalence or improvement. However, with the ability to manufacture multiple materials through additive manufacturing methods, traditional mechanical testing techniques can be extended to determine volume and interface performance, in order to better evaluate new additive manufacturing methods and material combinations [148,149]. In MMAM, the mechanical properties of parts are not only determined by the intrinsic characteristics of individual materials, but also jointly determined by key factors such as the metallurgical bonding quality at the interface of heterogeneous materials, the evolution of microstructure, and stress state. It is the core indicator for measuring the reliability of multi-material components in use. Different metal multi-material systems have

great differences in mechanical properties, especially in tensile properties. The following from the metal wire material additive manufacturing summarizes the mechanical properties under different material systems, as shown in Table 4. To provide theoretical support and technical reference for the design and optimization of high-performance MMAM components.

Table 4. A summary of tensile strength results for multi-material structures.

Material-1	Material-2	Methods	UTS(MPa)	Ref.
SS316L	IN625	GMAW	487	[150]
HSLA	IN625	WAAW	509	[151]
ER70S-6	ERNi-1	GTAW	634	[51]
ER70S-6	S355	WAAM	740	[92]
SS316L	ER70S-6	WAAM	502	[152]
ER120S-G	ERCuSi-A	CMT-WAAM	404	[153]
IN718	SS304L	CMT-WAAM	667	[154]
AA5052	AA7075	GTAW	182	[155]
Ti6Al4V	Al-6.21Cu	CMT-WAAM	164	[130]
NbZr1	Ti6Al4V	WAAM	567	[156]
Mg-Al-Si	Mg-Gd-Y-Zn	WAAM	236	[157]
P92	Steel-304H	WAAM	680	[158]
AISI 304	ErNiCu-7	GTAW	258	[159]
Cu	SS316L	CMT-WAAM	427	[160]
Al-6.5Cu	Al-6.4Mg	WAAM	258	[110]
Ti6Al4V	Al6061-T6	GTAW	230	[161]
Hastelloy C-276	SS316L	GTAW	780	[162]
AA8011	AA6061	CMT-WAAM	97	[163]
Al5356	Al6082	GMAW	263	[164]
NiTi	Nb	WAAM	789	[165]
AISI 310	9Cr-1Mo	GTAW	581	[166]
SS316L	Q345	GTAW	1073	[167]
SS316LSi	ER70S-6	WAAM	901	[168]
TA7	SS304	GTAW	293	[169]
AA5052	ERTi-2	WAAM	54	[170]
AA7075	AA6082	CMT-WAAM	212	[171]
AA5356	AA7075	EBWAM	289	[87]
ERNiCr-3	AISI 304L	WAAM	565	[172]
NiTi	Ti6Al4V	WAAM	460	[125]
IN601	ER304	GTAW	485	[173]

5. Conclusions and Prospects

5.1. Conclusions

This article reviews the latest progress, challenges, and potential of the MMAM process, and analyzes its application and the advantages and disadvantages of various processes. Metal multi-material wire additive manufacturing technology, through the precise control of heat sources such as electric arcs, lasers, and electron beams and the coordinated feeding of multiple wires, achieves the near-net forming of dissimilar metals and FGM, breaking through the limitations of traditional manufacturing in complex structures and multi-performance integration, and providing an innovative solution for the preparation of multi-functional components in aerospace and other fields.

In terms of the technical system, WAAM has become the core solution for manufacturing large multi-material components due to its low cost and high deposition efficiency. Its CMT variant inhibits the excessive growth of brittle IMCs through low heat input. LWAM relies on high-energy focusing to achieve a multi-material gradient transition for precision components. The multi-laser coaxial wire feeding technology enhances the ac-

curacy of composition control, laying the foundation for FGM's refined manufacturing. EBAM provides oxidation protection for active metal multi-material bonding in a vacuum environment, and its high energy density promotes metallurgical bonding at the interface of high-melting-point metals, expanding the possibility of manufacturing refractory metal composite structures.

Among the key process parameters, the wire feeding method directly determines the uniformity of material mixing and the continuity of the gradient. Multi-wire synchronous wire feeding enables real-time dynamic control of composition. The side-axis wire feeding, through multiple independent wire feeding devices, can directionally deposit dissimilar materials in specific areas. The WFS ratio affects the interface phase composition by controlling the proportion of molten pool components. The deposition sequence and the historical regulation of the current overheat cycle regulate the residual stress at the interface.

The interface control technology transition layer blocks the diffusion of harmful elements, alleviates thermal performance mismatch, and enhances the interface bonding strength. In the field of assistance technology, the ultrasonic field achieves grain refinement and interface bonding strength optimization through the cavitation effect and acoustic flow interaction. Post-processing technology optimizes performance through thermodynamic regulation.

MMAM technology demonstrates significant application potential across aerospace, energy, and medical fields. It enables the fabrication of multi-material lightweight structural components with integrated thermal management, overcoming the longstanding challenge of simultaneously achieving lightweighting, thermal management, and system integration. Furthermore, it facilitates the manufacturing of large-scale corrosion-resistant components with cladding or functional gradients for nuclear reactors, addressing the critical contradiction between material longevity and reactor safety in extreme environments. In the biomedical domain, this technology permits the production of bionic, multi-material orthopedic implants with site-specific properties, thereby resolving issues such as biomechanical mismatch, poor bio-fusion, and the lack of personalization inherent in traditional implants. This represents a paradigm shift from manufacturing homogeneous parts to designing and manufacturing integrated functional systems.

However, several key challenges impede its broad industrialization. First, insufficient interfacial wettability in high-melting-point differential systems can lead to incomplete fusion defects, restricting the composite application of high-temperature resistant and lightweight materials under extreme conditions. Second, the mismatch in thermal expansion coefficients between dissimilar materials generates interfacial shear stresses, which readily initiate microcracks and become a primary failure mechanism in components subjected to thermal cycling. Third, controlling the thickness of brittle IMCs layers and the coordinated optimization of functional performance remain critical bottlenecks. Breakthroughs in these areas rely on the deep integration of material design, process innovation, and cross-scale characterization techniques to propel the technology from laboratory research to industrial application. Essentially, the pathway to robust multi-material structures is not through a single innovation but through holistic integrated material design, in-process monitoring, and multi-scale modeling. The compiled research results provide a fundamental framework for understanding these complex interactions and lay the groundwork for the next stage of development.

5.2. Prospects

Metal multi-material wire additive manufacturing technology will develop in the direction of deep integration of intelligent regulation-multi-field collaboration innovation, promoting its transition from laboratory research to industrial mass production and providing core support for high-end equipment manufacturing.

In terms of intelligent and digital process optimization, the intelligent closed-loop control system is the core direction to break through the bottleneck of process stability. Through multi-dimensional monitoring such as infrared thermal imaging of the molten pool, plasma spectroscopy analysis, and high-speed photography, combined with the long short-term memory neural network algorithm, parameters such as WFS can be adaptively adjusted to reduce the interface defect rate.

In terms of deepening the multi-field collaborative interface regulation mechanism, multi-energy field coupling auxiliary technology achieves precise control of interface quality. Ultrasonic-magnetic field synergy: Ultrasonic vibration refines grains, and the alternating magnetic field enhances the convection of the melt, reducing the width of the dilution zone at the interface of the high-melting-point difference system. The laser–arc composite heat source optimizes energy input to resolve the contradiction between incomplete fusion and burn-off. Low-temperature plasma pretreatment technology improves the wettability of dissimilar materials and can reduce the interface contact angle.

In terms of new material systems and structural innovation, FGM and heterostructures are developing towards the coordinated optimization of performance–function–cost. The new transition layer material system breaks through the traditional limitations, and the biomimetic heterogeneous structure enables the component to have both high-temperature strength and low-temperature toughness. The integrated manufacturing of adaptive functional materials has become a new hot spot, enabling the development of intelligent structures and having application potential in fields such as aerospace.

In conclusion, through the deep integration of material design, process innovation, and intelligent control, this technology will gradually solve bottleneck problems such as interface brittleness and become the core manufacturing technology supporting the development of high-end equipment towards lightweight, integration, and intelligence.

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Abbreviations

The following abbreviations are used in this manuscript:

AM	Additive manufacturing
MMAM	Multi-material additive manufacturing
WAAM	Wire arc additive manufacturing
GMAW	Gas metal arc welding
GTAW	Gas tungsten arc welding
PAW	Plasma arc welding
CMT	Cold metal transfer
EBAM	Electron beam additive manufacturing
LWAM	Laser wire additive manufacturing
IMCs	Intermetallic compounds
DED	Directed energy deposition
FGM	Functionally graded material
UTS	Ultimate tensile strength
MWAAM	Multi-wire arc additive manufacturing
WLDED	Wire-feed laser directed energy deposition
WFS	Wire feeding speed
BS	Bimetallic structure
UPT	Ultrasonic peening treatment
PWHT	Post-weld heat treatment
YS	Yield strengths
TIG	Tungsten inert gas welding
MIG	Metal inert gas welding

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